When Child Benefits Arrive: Payment Timing, Liquidity, and Mothers' Welfare

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Abstract

The expanded 2021 Child Tax Credit raised benefits and paid half of them in monthly advances. Holding each family's annual credit fixed, we ask what the earlier payments were worth to married and single mothers in the Survey of Income and Program Participation. Relative to comparable women without children, mothers show no detectable change in employment and no detectable relative increase in bank balances; card debt fell, mostly among mothers who held some cash and carried debt, while low cash persisted among single mothers. A monthly consumption-saving model with income and childcare-cost risk, in which some mothers save and others spend all their cash, is estimated by maximum likelihood on each mother's records with patience set at a standard value. The model-based spender probability averages 31 percent for single mothers and 2 percent for married mothers. Relative to paying the same credit at filing, the 2021 calendar was worth $191 per married mother and $987 per single mother, and $993 and $1,820 for mothers with $1,000 or less in liquid assets. For the average married and single mother, each added credit dollar paid only at filing was worth 92 and 60 cents, against 102 and 96 cents paid as in 2021. The single-mother value is conditional on the spender description: valuing every mother as a saver gives $476, and impatient savers with conventional discount factors, valued at patient rates, give $483 to $539, about half as large.

Keywords: Child Tax Credit; payment timing; liquidity; consumer credit; single mothers; household welfare.
JEL: D14, D15, H24, H31, I38.

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1 Introduction

The American Rescue Plan Act made the 2021 Child Tax Credit (CTC) larger and paid half of it early. The maximum credit rose from $2,000 per child to $3,600 for each child under six and $3,000 for each child aged six to seventeen, and the credit became fully refundable. From July 15 to December 15, 2021, the Internal Revenue Service paid half of each family's estimated credit in six monthly advances of up to $300 per young child and $250 per older child; the remainder came with 2021 tax returns in spring 2022 (Internal Revenue Service, 2021, 2022). The expansion lasted one year. From tax year 2022 the credit returned to $2,000 per child, paid once a year at filing.

Whether child benefits should be paid monthly or at tax time remains an open design question, and proposals to restore the expansion would make monthly payment permanent (DeLauro et al., 2023). The 2021 reform, however, changed the amount and the timing of the credit at the same time, so the evidence that hardship fell while the advances were paid (Parolin et al., 2021; Shafer et al., 2022) does not say how much of the gain came from receiving the money earlier. This paper asks three questions about the earlier payments. Who was affected by them? Why did their timing matter? And how much was earlier payment worth, holding each family's annual credit fixed?

We answer them for married and single mothers in the Survey of Income and Program Participation (SIPP), which records each mother's monthly income and employment, whether and how much her family paid for childcare in each year, and her household's bank balances and credit-card debt at the end of 2020 and 2021, and for part of the sample 2019. The two groups differ in exactly the ways that matter for timing. The 593 married mothers have a median 2020 family income of $116,000, and one in nine had $1,000 or less in liquid assets at the end of 2020; we call such mothers cash-poor. The 147 single mothers in the model sample have a median income of $31,000, and nearly half had $1,000 or less; the descriptive and reduced-form results use all 159 single mothers who meet the sample rules.

Who was affected. Full refundability gave the largest increases to mothers with low earnings: the added credit averaged $2,758 for single mothers and $1,958 for married mothers. The mothers who gained most also held the least cash; for those with $1,000 or less in liquid assets, the added credit was 13 times the median bank balance for married mothers and 28 times for single mothers. Relative to women of the same marital status without children at home, we find no detectable change in employment rates when the advances began, in the SIPP or in matched monthly Current Population Survey (CPS) data, no change in the use of paid childcare, and no detectable relative increase in bank balances, although the bank estimates are too imprecise to rule out substantial saving. Card debt fell relative to comparison women: by $1,909 (SE 681) for married mothers, and by $1,010 (SE 469) for single mothers who carried card debt.

Why. Splitting these responses by starting cash shows two patterns, although few differences between cash groups are statistically distinguishable. The fall in card debt is concentrated among mothers with some cash who carried card debt, the response one would expect from households facing a 16 percent borrowing rate. Among single mothers with little cash, those who carried card debt reduced it, those without started borrowing at about the same rate as comparison women, and bank balances showed no detectable relative increase; spending data show that families with the smallest buffers spent 73 cents of each advance dollar in its first week (Wheat et al., 2022). Low cash is also persistent among single mothers: 81 percent of those with $1,000 or less in December 2019 were still that low a year later, against 47 percent of married mothers. Parolin et al. (2023) find

that the monthly payments and the lump sum at filing reduced different hardships; our evidence indicates that the responses to the advances differed by cash on hand, and that mothers differ in how they hold it. The model allows two kinds of mothers, savers who keep buffers and spenders who consume whatever cash they have, a parsimonious description that the December balances fit better than savers alone. The fitted spender probability, a model-based classification rather than an observed behavior, averages 31 percent (SE 6) for single mothers and 2 percent (SE 2) for married mothers, and is concentrated among mothers who started 2021 with little cash.

How much. To value earlier payment, we build a monthly buffer-stock consumption-saving model with income risk and a childcare-cost process, solved for each mother from her own resources, credits, childcare costs and starting cash. Savers choose how much cash to carry; spenders consume it all each month, the limit of strong present bias. We set the annual discount factor at 0.96, a standard value in buffer-stock models (Carroll, 1997), because December balances can show that a mother keeps no buffer but not how impatient she is; the spender share takes the place of a low discount factor and is identified by the balances. Income risk is estimated by maximum likelihood from each mother's monthly income records, the childcare process from her annual childcare indicators and fees from her reported weekly payments, separately for married and single mothers, and the spender shares from December 2021 balances, with design-based standard errors that include first-step uncertainty. We value each payment calendar by the grant, spread evenly over July 2021 to June 2022, that would make a mother equally well off under old-law payment, and we value spenders' consumption with the same discount factor as savers'. Holding the annual credit fixed, the 2021 calendar was worth $191 (SE 37) per married mother and $987 (SE 127) per single mother, compared with paying the same credit entirely at filing. Paid only at filing, each added dollar of the expanded credit would have been worth 92 cents to the average married mother and 60 cents to the average single mother; paid as in 2021, 102 and 96 cents, more than a dollar for married mothers because the advances also paid half of the old-law credit before filing. The calendar was thus worth about 10 and 36 cents per added dollar. The value is concentrated among mothers with little cash: $993 for married and $1,820 for single mothers with $1,000 or less in liquid assets, against $6 and $28 for mothers with $25,000 or more. Within income groups, low cash still predicts large gains, in large part because the spender probability rises as cash falls.

Savers alone value the 2021 calendar at $152 (married) and $476 (single); spenders, for whom six monthly payments replace one lump, at $1,132 and $1,824. The single-mother values depend more on how cash-poor mothers are modeled than on patience, how balances are measured or starting conditions: their averages change by at most 9 percent for discount factors between 0.90 and 0.99, with each mother's reported advance receipts in place of the statutory schedule in the likelihood, without the outlier component of the reporting error, with part of the spring 2021 stimulus added to starting cash, or with July childcare status set from the chain. They change by up to 12 percent when the refundable credits of tax year 2020 are added to starting cash. The married average is more sensitive to patience, risk aversion and withholding, ranging from $112 to $408. But if cash-poor mothers saved a little, as an impatient saver at a conventional discount factor would, and were valued at savers' patience, single mothers' average value would be $483 to $539, about half as large; that model fits the balances worse (Section 9.5). Counting spenders' consumption with their own impatience, rather than at patient rates, would raise the values. Higher risk aversion, which fits the balances better, also raises them; valuing the advances mothers report receiving, a third short of the schedule for single mothers, lowers them by a quarter to a third; and the 2021 design made permanent would be worth $58 (married) and $748 (single) a year when the two long-run states are compared, and $153 and $813 counting the one-time release of the buffers savers no longer need

(Section 9.5).

A single July check would have been worth less than the six advances, because spenders would have consumed it at once. Borrowing at credit-card rates substitutes partly for earlier payment for savers and much less for spenders, who pay interest on what they borrow: with three extra months of card credit, cash-poor single mothers still value the 2021 calendar at $1,557.

Contribution. The paper connects three literatures. Studies of the expanded credit document effects on hardship, consumption, financial distress and employment (Parolin et al., 2021, 2023, 2024; Shafer et al., 2022; Bovell-Ammon et al., 2022; Karpman et al., 2022; Hamilton et al., 2022; Pilkauskas et al., 2024; Batra et al., 2023; Schild et al., 2026; Ananat et al., 2024; Enriquez et al., 2023; Pac and Berger, 2024). Parolin et al. (2023) compare the monthly and lump-sum parts of the calendar directly, and Lourie et al. (2026), with transaction data on nearly two million individuals, show that spending rose in the week after each monthly payment and that overdrafts and payday-loan use fell among lower-income recipients, more than after annual tax refunds. These studies measure responses to the payments. We add a dollar value of earlier payment, separated from the larger entitlement and computed for mothers with different amounts of cash, which no comparison in the data can provide; Wheat et al. (2022) already show that families with small buffers spent more of each advance. A second literature shows that spending and food intake follow the timing of benefits in SNAP, Social Security and stimulus payments (Wilde and Ranney, 2000; Todd, 2015; Goldin et al., 2022; Shapiro, 2005; Stephens, 2003; Hastings and Washington, 2010; Parker, 2017; Johnson et al., 2006; Parker et al., 2013), that other outcomes, such as crime, follow SNAP disbursement dates (Carr and Packham, 2019), that households spend SNAP benefits differently from cash (Hastings and Shapiro, 2018), and, for the EITC, that household debt follows the date of the credit (Jones and Michelmore, 2019). The evidence on delivery is mixed: Aguila et al. (2017) find smoother consumption when benefits are paid more often, few recipients chose the Advance EITC (Jones, 2010), Sahm et al. (2012) and Caldwell et al. (2023) study withholding and refund uncertainty, and for the expanded credit Sisha et al. (2025) find larger short-run improvements in reported mental health after the lump sum at filing than after the monthly payments. A third models buffer-stock saving with borrowing constraints (Deaton, 1991; Carroll, 1997) and documents that many households hold little liquid wealth persistently (Kaplan et al., 2014; Aguiar et al., 2025). We estimate such a model, with savers and spenders (Campbell and Mankiw, 1989), on individual records for two groups of mothers and use it to value a specific policy calendar. Households may also treat expected refunds and their regular payments differently (Baugh et al., 2021), which the savers-and-spenders description leaves open.

Section 2 describes the policy and the compared calendars, and Section 3 the SIPP samples. Section 4 asks who was affected and Section 5 why. Sections 6 to 8 develop, estimate and assess the model, Section 9 reports the value of earlier payment, Section 10 adds credit-card borrowing, and Section 11 concludes.

2 The 2021 credit and the compared payment calendars

2.1 What changed in 2021

Under prior law, the CTC was $2,000 per child under seventeen, partially refundable, and paid with the annual tax return. Families with little or no earnings received little of it: the refundable part

was limited to 15 percent of earnings above $2,500 and to $1,400 per child. The American Rescue Plan Act changed three things for tax year 2021. It raised the maximum to $3,600 for each child under six and $3,000 for each child aged six to seventeen, with the increase above $2,000 phased out above $150,000 of income for married couples filing jointly and $112,500 for heads of household. It made the credit fully refundable. And it paid half of each family's estimated credit in advance, in six monthly payments from July 15 to December 15, 2021, with the remainder claimed on the 2021 return in spring 2022 (Internal Revenue Service, 2021, 2022). From tax year 2022 the credit reverted to prior law.

Figure 1 shows the calendar for a family with one child under six. Under prior law the family would have received $2,000 at filing in spring 2022. Under the 2021 law it received $300 a month from July to December 2021 and $1,800 at filing, a total of $3,600. From 2023 the family again receives $2,000 a year at filing.

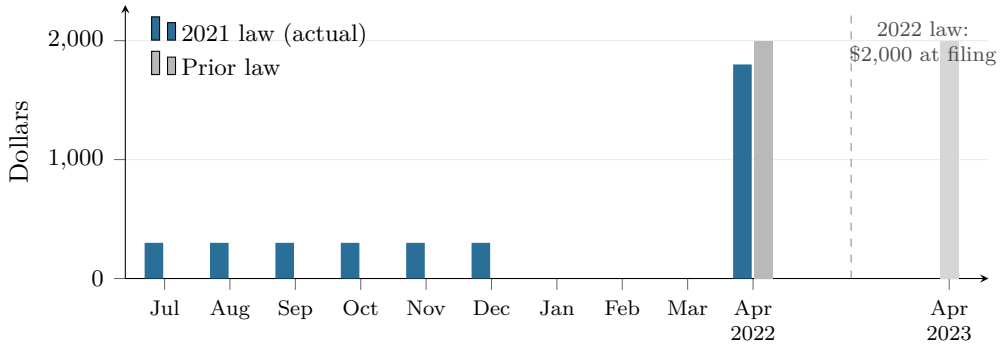


Figure 1: CTC payments for a family with one child under six. Monthly advances July–December 2021 (2021 law) and the filing payment for tax year 2021 in April 2022, against prior law. Right of the dashed line: the filing payment for tax year 2022, when the credit returned to prior law.

2.2 Calendars compared in the model

Let B_i^O and B_i^N denote mother i's old-law and expanded annual credits for tax year 2021, computed from her family's 2021 earnings, other income, children's ages and childcare expenses. Month $t = 0$ is July 2021, and the filing payment is placed in April 2022 ($t = 9$). Table 1 lists the calendars. The amount comparison changes the credit while keeping filing-only payment (O against F). The timing comparison keeps the expanded credit and changes only when it is paid (F against S). Two further calendars ask whether paying the advanced half at once (U) or advancing the whole credit (M) would have added value. Other refundable credits (the earned income tax credit and the child and dependent care credit) arrive in the same April under every calendar, and after June 2022 every calendar continues with the same 2022-law year, repeated: ordinary credits each April, with the same income and childcare risks.

Table 1: First-year CTC calendars

Calendar	July–December 2021	April 2022 filing
O: old law at filing	nothing	B_i^O
F: expanded credit at filing	nothing	B_i^N
S: 2021 law (actual)	$B_i^N/12$ each month	$B_i^N/2$
U: advanced half in one July check	$B_i^N/2$ in July	$B_i^N/2$
M: whole credit in six advances	$B_i^N/6$ each month	nothing

Eligibility, earnings and family composition are held fixed across calendars. The comparison is therefore between payment schedules for given annual entitlements, not a full counterfactual of the American Rescue Plan Act, which also expanded the earned income and childcare credits and paid a third round of stimulus checks in spring 2021.

3 Data

3.1 The SIPP samples

The SIPP is a longitudinal survey that records monthly employment, earnings, family income and program receipt over each reference year, together with household asset and debt balances at the end of the year. We use the 2021 and 2022 releases, which cover reference years 2020 and 2021, and the 2020 release for a placebo year. Respondents in the 2020 and 2021 panels are observed at the end of 2020 and again through 2021, so each mother's December 2020 bank balance precedes the advances and her December 2021 balance follows them. The 2022 release also records, by month, whether the family received an advance payment and its amount.

We construct two samples with one set of rules (Appendix A). A *married mother* has her spouse present and a valid spouse link in all twelve months of 2021, is the householder or the householder's spouse, and was also a married mother in December 2020. A *single mother* is unmarried (widowed, divorced, separated or never married) in all twelve months of 2021 and in December 2020, and is the householder. In both samples the mother is aged 18–54, has at least one own child under eighteen in the household in every month of 2021 and in December 2020, reports valid earnings and family income in all twelve months, is in the universe of the SIPP childcare questions, and has a December 2020 bank record. We also apply the screens of the original welfare sample: known advance and stimulus receipt in every month, reported advances no larger than the computed credit, and positive resources after childcare costs.

These rules give 593 married mothers and 159 single mothers. For the model we additionally require single mothers to have at least \$500 a month of measured cash resources excluding refundable credits. This drops 12 mothers in deep poverty, whose 2021 credit was up to 125 times their annual measured resources and whose consumption the model cannot represent without data on in-kind support. The married sample's lowest value is \$435 a month, so the restriction brings the two samples' lower supports close together. The model sample of single mothers has 147 mothers. Descriptive and reduced-form results use all 159. Restricted to the 147, the decline in card debt is similar for single mothers who carried card debt (\$1,092 against \$1,010) and larger, but still imprecise, for all single mothers (\$549 against \$360; Appendix Table 24). Excluding the poorest mothers, who gained most from full refundability, likely lowers the average single-mother welfare values.

3.2 Measurement

Resources and credits. Annual resources are the sum of the family's monthly incomes in 2021, split into the mother's earnings, her spouse's earnings, and other income. Taxes and credits are computed for each mother from these components, her children's ages and her childcare expenses: under 2021 law, under prior law, and under 2022 law for the years after the expansion. Married mothers file jointly and single mothers as heads of household. For married mothers we reproduce the TAXSIM-based inputs of the original sample exactly; for both samples the calculations use the Tax-Calculator microsimulation model with the same credit formulas, which matches TAXSIM to within \$127 for every married mother and within \$5 for 94 percent of them (Appendix A). We denote by μ_i a mother's monthly resources after federal income and payroll taxes, excluding the CTC and other refundable credits.

Liquid assets and debt. Liquid assets are the household's bank balances, stocks and mutual funds, and bonds in December 2020. Groups by liquid assets ($\leq$\$1,000, \$1,000–25,000, $\geq$\$25,000) are fixed at that date. Credit-card debt is the household's reported card balance in December of each year. The model's saving state is the December 2020 bank balance. In the text, cash-poor mothers are those with \$1,000 or less in liquid assets in December 2020, and cash in the model is the bank balance.

Childcare. Paid childcare status and the weekly payment come from the SIPP childcare questions for reference years 2020 and 2021. A mother pays for childcare in a year if she reports a positive payment for any arrangement. We classify mothers by the age of their youngest child at the end of 2021.

Weights and inference. We use the December 2021 person weights and the SIPP's 240 Fay replicate weights; all standard errors are design-based, $\widehat{\mathrm{Var}}(\hat{\theta}) = \frac{4}{240}\sum_r(\hat{\theta}_r - \hat{\theta})^2$.

3.3 Comparison groups

Two reduced-form designs compare mothers with women of the same marital status who had no own children under eighteen at home in 2021 or in December 2020, selected by the same rules apart from those on children and childcare. There are 299 married and 417 single comparison women. They received no advances, but they did receive the third stimulus payment in spring 2021, which we control for.

3.4 Descriptive statistics

Table 2 describes the two samples. Married mothers are older and richer. In the model sample of 147, single mothers have a median 2020 family income of \$31,000 (\$28,000 among all 159), against \$116,000 for married mothers, and 69 percent of them have family income below \$50,000. They hold far less cash: the median single mother had \$1,100 in liquid assets at the end of 2020, against \$19,000 for married mothers, and 48 percent had \$1,000 or less. Single mothers are more likely to work and slightly more likely to pay for childcare. Their weekly childcare payments are about half those of married mothers but about 1.6 times as large relative to resources (the ratio of the medians in Table 2; the fee shares estimated in Section 7 are more than twice as large). The added credit from the expansion is larger for single mothers, a median of \$2,100 against \$1,600 (means \$2,758 and \$1,958), because full refundability raised the credit most for families with low earnings; half of single mothers receive the earned income tax credit, against one in ten married mothers. One

in five single mothers lives with an unmarried partner, whose income is not part of SIPP family income, and about one in five lives with other adult relatives, whose income is.

Table 2: Descriptive statistics, SIPP mothers

	Married	Single mothers	
	mothers	All	Model sample
Mothers	593	159	147
Age, median	38	37	37
Children under 18, median	2	2	2
Has a child under 6 (%)	49.3	41.9	41.3
2020 family income, median ($)	116,024	28,450	30,960
2020 family income below $50,000 (%)	15.9	71.4	68.9
Months employed in 2021 (%)	64.1	81.0	85.8
Monthly resources excluding credits μ_i, median ($)	8,749	2,816	2,992
Bank balance, December 2020, median ($)	15,000	1,000	1,100
Liquid assets, December 2020, median ($)	19,000	1,000	1,100
Liquid assets ≤$1,000 (%)	10.8	50.7	47.9
Any credit-card debt, December 2020 (%)	41.0	52.9	55.9
Credit-card debt, December 2020, median among debtors ($)	4,200	3,000	3,000
Paid childcare in 2021 (%)	24.6	27.9	30.3
Weekly childcare payment, median among payers ($)	200	110	110
Expanded 2021 CTC, median ($)	5,800	6,000	6,000
Prior-law CTC, median ($)	4,000	2,000	2,000
Added credit, median ($)	1,600	2,200	2,100
Advances reported, July–December 2021, median ($)	1,800	1,500	1,500
Receives EITC, tax year 2021 (%)	10.2	51.0	50.6
EITC, median among recipients ($)	2,893	2,623	2,768
Unmarried partner in household (%)	0.2	21.6	22.6
Other adult relatives in household (%)	16.4	21.5	23.3

Weighted by December 2021 person weights; medians are weighted medians. The model sample of single mothers excludes 12 with monthly resources below $500. Means of the credits and advances, which the welfare calculations use, are in Tables 3 and 12. Liquid assets: bank, stocks and mutual funds, and bonds. Credits computed with 2021-law and prior-law rules for each mother's 2021 incomes and children.

4 Who was affected?

4.1 Who received how much

The expansion raised the credit most for mothers with low earnings, because it made the credit fully refundable. In this section, the tables of cash and credits use the model samples (593 married and 147 single mothers); the comparisons with women without children use all 159 single mothers. Single mothers gained $2,758 on average, against $1,958 for married mothers (Table 3), and among single mothers with 2020 family income below $25,000 the added credit averaged $3,853.

The same mothers held the least cash. Table 3 compares each group's December 2020 bank balance with the credit it gained. For the median married mother, the added credit was a small fraction of her bank balance. For married mothers with $1,000 or less in liquid assets, the added credit of $3,063 was 13 times their median bank balance of $230, and a year later 55 percent of them still

had $1,000 or less in the bank. Nearly half of single mothers had $1,000 or less in liquid assets; for them the added credit of $3,375 was 28 times the median balance of $120, and two thirds still had $1,000 or less in the bank a year later. For these mothers, waiting until tax time for the credit means doing without it in the meantime.

Table 3: Cash on hand and the credit

	Married mothers		Single mothers	
	All	Liquid ≤$1k	All	Liquid ≤$1k
Mothers	593	71	147	72
Share of group (%)	100	10.8	100	47.9
Bank balance, December 2020, median ($)	15,000	230	1,100	120
Added annual credit, mean ($)	1,958	3,063	2,758	3,375
Monthly advance, mean ($)	470	567	457	497
Bank ≤$1,000 in December 2021 (%)	10.0	54.7	45.0	66.7

Model samples. Liquid assets in December 2020. Added credit: expanded 2021 credit minus prior-law credit. Monthly advance: one twelfth of the expanded credit.

4.2 What mothers did with the money

We estimate a two-period difference-in-differences, comparing the change from December 2020 to December 2021 for mothers with the change for women of the same marital status without children at home, who received no advances:

$$\Delta y_i = \psi_0 + \psi \, \text{Mother}_i + X_i'\pi + \eta_i, \tag{1}$$

where Δy_i is the change in household card debt (in dollars, winsorized at the 1st and 99th percentiles), an indicator of any card debt, the inverse hyperbolic sine of card debt, or the bank balance. The controls X_i are December 2020 card debt, bank balance and liquid assets (each in inverse hyperbolic sine, with squares), an indicator of liquid assets at or below $1,000, age and its square, log family income, and the amount of stimulus payments received in 2021. Controlling for starting debt matters because high balances tend to fall: the comparison women started with more card debt. Appendix Table 21 compares the two groups in December 2020. Comparison women are older, by three years among married and five among cash-poor single women, and received smaller stimulus payments; single comparison women also had higher incomes and more cash, but within the group with $1,000 or less their median income and bank balance are close to the mothers'. The mechanism results in Section 5 therefore compare women within the same cash group.

Card debt fell. Table 4 reports the results. Married mothers reduced card debt by $1,909 (SE 681) relative to married women without children, and by $2,745 among mothers who carried card debt at the end of 2020. The average married mother was scheduled to receive $2,819 in advances and reported receiving $2,002, so the estimate is about two thirds of the scheduled advances, with a wide confidence interval. Because annual card balances also reflect new borrowing, we read this as evidence that card debt fell substantially, not as the share of the advances used for repayment. Among women observed in December 2019, 2020 and 2021, married mothers show no difference in the year before the advances: $606 (SE 933) from December 2019 to December 2020. The placebo is imprecise: its 95 percent interval, −$1,223 to $2,435, rules out a decline as large as the 2021 estimate but not one half its size. Across specifications that control for starting debt, the estimate

ranges from $1,135 to $2,082 (Appendix Table 24). Single mothers hold less card debt, and the average effect is smaller and imprecise (−$360, SE 292). Among single mothers who carried card debt, the reduction is $1,010 (SE 469), the share still carrying any card debt fell by 20 points (SE 8.7), and the balance fell by 38 log points (SE 15). The placebo change in 2020 is small (−$147, SE 568), though it rests on only 61 single mothers observed in all three years.

Table 4: Change in credit-card debt, December 2020 to December 2021: mothers minus comparison women

	Married mothers	Single mothers
Change in card debt ($)		
All mothers	−1,909 (681)	−360 (292)
Mothers with card debt in December 2020	−2,745 (1,251)	−1,010 (469)
Mothers without card debt in December 2020	−1,032 (750)	295 (300)
Change in having any card debt (percentage points)		
All mothers	−4.7 (3.7)	−10.8 (6.4)
Mothers with card debt in December 2020	−6.1 (5.4)	−20.1 (8.7)
Change in asinh(card debt) (log points)		
All mothers	−0.20 (0.10)	−0.14 (0.09)
Mothers with card debt in December 2020	−0.31 (0.16)	−0.38 (0.15)
Placebo: women observed in December 2019, 2020 and 2021		
Change 2019 to 2020 ($)	606 (933)	−147 (568)
Change 2020 to 2021, same women ($)	−2,319 (1,354)	−574 (448)
Mothers / comparison women	593 / 299	159 / 417
Placebo sample: mothers / comparison women	250 / 125	61 / 204

Coefficients on the mother indicator in equation (1); rows by December 2020 card-debt status interact the indicator with that status. Comparison women: same marital status, no own children under 18 at home. Standard errors from 240 Fay replicate weights.

No detectable relative increase in bank balances. The mother-minus-comparison change in bank balances is $3,402 (SE 6,779) for married mothers and −$3,035 (SE 3,199) for single mothers. For married mothers we also find no difference in the change in log liquid assets (0.09, SE 0.11) or in the share with $1,000 or less (Appendix D). These estimates are imprecise: they cannot rule out that a substantial part of the advances was saved.

No detectable change in employment. Figure 2 plots the monthly employment rate of SIPP mothers minus comparison women in 2021, relative to June. There is no break in July for either group. For married mothers, employment in July–December relative to January–June is 1.9 points lower (SE 1.4), and 0.0 (SE 2.0) after removing the pre-July trend. For single mothers the change is 2.1 points (SE 1.8), or 1.7 (SE 4.7) net of trend. In the matched monthly CPS, which follows many more women and both switches in delivery, the change in the monthly entry rate into work when advances began was −0.64 points (SE 0.63) for married mothers and −0.30 (SE 0.96) for single mothers, and the change in employment was 0.61 (SE 0.91) and −0.14 (SE 0.98) (Appendix C). Prior studies also find little change in parental employment (Ananat et al., 2024; Enriquez et al., 2023; Pac and Berger, 2024; Pilkauskas et al., 2024), although simulations with standard labor-supply elasticities predicted that replacing the old credit with a fully refundable allowance would lead about 1.5 million working parents to stop working (Corinth et al., 2021). Some contrasts in work entry are significant, notably a rise in single mothers' entry in January and February 2022 (Section 5). These estimates describe average employment; holding each mother's work fixed, as

the model does, is a maintained assumption that they do not test, and Section 9.5 values mothers who entered work in 2021 without their new earnings.

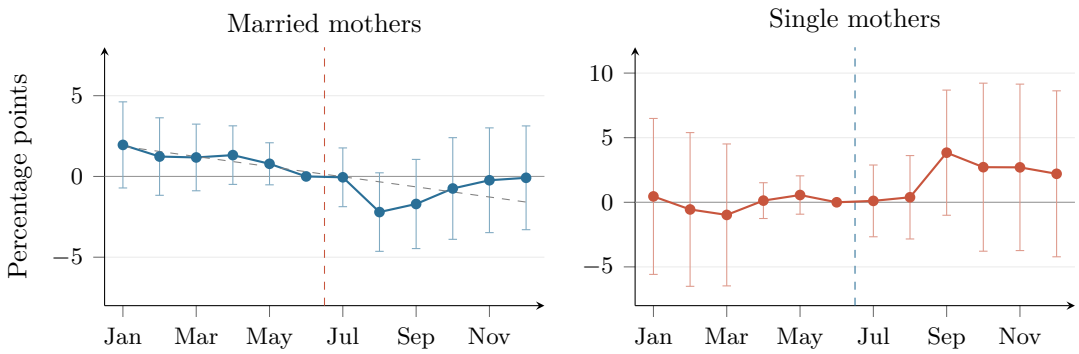


Figure 2: Employment of SIPP mothers minus women of the same marital status without children at home, 2021, relative to June. Any job in the month. Bars: 95% confidence intervals from replicate weights. Dashed line (married): linear pre-July trend, extended. Vertical line: first advance.

Childcare use barely moved. The share of mothers paying for childcare was 24.5 percent in 2020 and 24.6 percent in 2021 among married mothers, and 28.7 and 30.3 percent among single mothers. Among married mothers who did not work, larger advances did not raise the probability of paying for care (−4.0 points per $1,000 of advance, SE 2.9).

5 Why were they affected?

Three explanations could make the timing of the credit matter. Mothers may be unable to borrow cheaply against a credit they know is coming (liquidity constraints). They may discount the future heavily and so keep little cash (impatience). Or they may face large, unpredictable expenses and income drops that a small buffer cannot absorb (risk). The first and third imply that earlier payment improves mothers' lives by smoothing consumption; the second implies that it moves spending forward.

5.1 Spending and hardship followed the payments

The SIPP does not measure consumption, but bank-account and survey data do. Figure 3 reproduces the JPMorgan Chase Institute's comparison of weekly spending by families that received the first advance on July 15, 2021, with families that did not (Wheat et al., 2022). Spending was flat relative to non-recipients before the payment and, in the week it arrived, rose by about 37 percent of the payment, as read from their figure. Over the first week, recipients spent about 40 cents of each advance dollar, and families with the smallest cash buffers spent 73 cents. A household that could borrow freely against the coming credit would not change its spending on the day the money arrives. Lourie et al. (2026) find the same pattern in transaction data on nearly two million individuals: spending rose in the week after each monthly payment, most among the lowest-income recipients and those with more children, and overdrafts, payday-loan use and gig work for supplemental income fell among lower-income recipients.

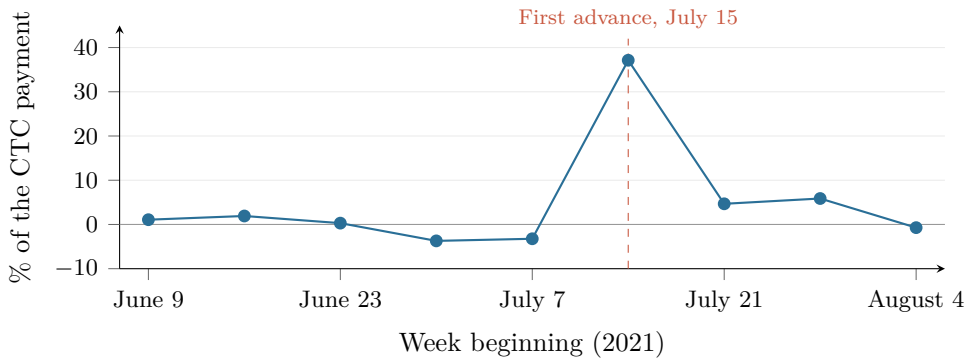


Figure 3: Weekly spending of families that received the first advance minus families that did not, as a percentage of the payment. Chase account holders; points digitized from Wheat et al. (2022), Figure 2.

Studies of material hardship show that it moved with the payments (Parolin et al., 2021; Shafer et al., 2022; Bovell-Ammon et al., 2022; Parolin et al., 2023). Using the Household Pulse Survey, Parolin et al. (2021) and Shafer et al. (2022) find that food insufficiency among families with children fell by about a quarter after the first advance, concentrated among low-income families. When the monthly payments stopped, food insufficiency rose by about a quarter, significantly only in the survey waves from late April 2022, after the filing season, with a large rise among low-income households (Bovell-Ammon et al., 2022). Parolin et al. (2023) compare the two parts of the calendar directly: the monthly payments reduced food insufficiency, while the lump sum at filing reduced the share of families behind on housing payments. Surveys find larger declines in food insecurity among recipients, less reliance on credit cards and high-cost credit, and better ability to pay bills (Karpman et al., 2022; Hamilton et al., 2022; Pilkauskas et al., 2024; Rook et al., 2023), along with fewer symptoms of depression and anxiety (Batra et al., 2023); Sisha et al. (2025) find larger short-run improvements in these symptoms after the lump sum at filing than after the monthly payments. In the Consumer Expenditure Survey, families spent about 44 cents of each dollar, mostly on housing and food (Schild et al., 2026). These studies do not separate the larger amount from earlier payment, and because the Household Pulse Survey does not record cash on hand, they cannot show whether the effects run through liquidity.

5.2 Liquidity: who repaid debt and who did not

If mothers were constrained by cash, how they used the advances should depend on how much cash they had. Table 5 splits the card-debt regression (1) by December 2020 liquid assets and card debt, and by liquid assets and income.

Table 5: Card debt and bank balances by starting cash: mothers minus comparison women

	Married mothers		Single mothers	
	Card debt ($)	Bank ≤$1k (pts)	Card debt ($)	Bank ≤$1k (pts)
By liquid assets and card debt, December 2020				
Liquid ≤$1k, card debt	−2,612 (8,197)	14.8 (20.2)	−401 (832)	−6.4 (11.4)
Liquid ≤$1k, no card debt	805 (909)	−7.2 (14.0)	991 (438)	10.0 (8.0)
Liquid >$1k, card debt	−2,889 (1,222)	−3.1 (3.7)	−1,395 (581)	18.7 (9.6)
Liquid >$1k, no card debt	−1,164 (776)	4.0 (2.2)	−232 (321)	17.1 (8.3)
By liquid assets and 2020 income (married: $50k; single: $25k)				
Liquid ≤$1k, lower income	2,029 (1,581)	−5.2 (14.3)	1,411 (618)	8.2 (7.8)
Liquid ≤$1k, higher income	−3,062 (5,101)	3.1 (19.2)	−702 (745)	−6.2 (12.6)
Liquid >$1k, lower income	98 (953)	5.6 (8.8)	−188 (565)	45.6 (14.5)
Liquid >$1k, higher income	−2,238 (704)	0.0 (1.9)	−1,110 (428)	11.7 (6.1)

Coefficients on the mother indicator interacted with each group in equation (1), with group main effects. Card debt: change December 2020 to December 2021 (dollars, winsorized). Bank ≤$1k: indicator of a bank balance of $1,000 or less in December 2021 (percentage points). Replicate standard errors. Married mothers: 593 (comparison 299); single mothers: 159 (comparison 417). Cells with liquid assets ≤$1k contain 28–49 mothers.

Two patterns stand out. First, the decline in card debt is concentrated among mothers who held some cash and carried card debt: $2,889 (SE 1,222) among married and $1,395 (SE 581) among single mothers in that group, with the share of single mothers carrying any debt falling by 24 points (SE 11). A mother with a buffer who pays 16 percent on her card balance gains by using an earlier payment to reduce it; this is the behavior of a patient household that faces a gap between the borrowing and the saving rate.

Second, among mothers with little cash, card debt did not detectably fall relative to comparison women in the main estimates, and bank balances showed no detectable relative increase. Table 6 looks inside these cells. Cash-poor single mothers who carried card debt reduced it at least as much as comparison women: their mean card debt fell from $4,467 to $2,395, 42 percent paid it off, against 30 percent of comparison women, and with changes capped at $5,000 the relative decline is $1,001 (SE 499). Among those without card debt, the same share started borrowing during 2021 as among comparison women, 19 percent, but those who did took on larger card balances. The positive estimate of $991 (SE 438) for this cell rests on about ten new card balances; with changes capped at $5,000 it is $463 (SE 301). The positive estimate of $1,411 (SE 618) for lower-income cash-poor single mothers comes mostly from comparison women with card debt, who started with balances nearly three times as large and cut them by about $4,300; capped, it is $349 (SE 333).

The estimates for cash-poor married mothers are too imprecise to interpret, with 7 and 16 comparison women in the two cells. Cash-poor single mothers without card debt were also 10 points (SE 8) more likely than comparison women to end 2021 with $1,000 or less in the bank, and single mothers with some cash but low income, a group of only 13 mothers, 46 points (SE 15).

Few of the differences between cash groups are statistically distinguishable. Of the 16 differences in Table 5 between mothers with $1,000 or less and those with more, within card-debt status or income group, two are significant at 5 percent, and neither survives a Bonferroni correction for the number of tests. Among single mothers without card debt, the card-debt estimate is $1,223 higher for those with little cash (SE 507, $p = 0.02$; married mothers, $1,969, SE 1,076, $p = 0.07$), but this rests on the ten or so new card balances discussed above: with changes capped at $5,000

the difference is $524 (SE 325, $p = 0.11$). Among lower-income single mothers, the bank-balance estimate is 37 points lower for those with little cash (SE 16, $p = 0.02$), because the other cell, 13 mothers with some cash, is the one whose balances fell. The card-debt estimates for mothers who carried debt do not differ significantly by cash (married: $277, SE 8,193; single: $994, SE 1,062). Part of the split by starting debt is accounting: a mother without card debt cannot reduce her own balance, so for her groups the estimates show only whether mothers started borrowing more or less than comparison women.

Table 6: Card debt by December 2020 cash and card debt

A. Card debt: levels and changes	N	Mothers' card debt	Change in card debt ($)		Paid off or
		mean, 2020 → 2021	Mothers	Comparison	started (%)
Married mothers					
>$1k, card debt	216 / 134	8,991 → 6,933	−1,767 (903)	−450 (1,201)	31.2 / 22.6
>$1k, no card debt	306 / 142	0 → 1,162	1,132 (172)	2,179 (686)	24.3 / 24.4
≤$1k, card debt	28 / 7	10,087 → 10,374	288 (2,164)	−465 (8,144)	18.2 / 11.7
≤$1k, no card debt	43 / 16	0 → 1,283	1,189 (768)	336 (237)	15.7 / 19.1
Single mothers					
>$1k, card debt	38 / 139	4,685 → 2,260	−2,425 (559)	−1,857 (730)	53.9 / 31.4
>$1k, no card debt	39 / 165	0 → 274	274 (109)	547 (171)	20.2 / 17.3
≤$1k, card debt	35 / 45	4,467 → 2,395	−2,072 (840)	−2,456 (1,197)	41.9 / 30.1
≤$1k, no card debt	47 / 68	0 → 799	799 (358)	316 (114)	19.5 / 19.4

B. Mothers minus comparison women	Main ($)	Capped ±$5k ($)	Any debt (pts)	asinh	Share (pts)
Married mothers					
>$1k, card debt	−2,889 (1,222)	−1,101 (401)	−7.2 (5.8)	−0.35 (0.17)	8.6 (5.2)
>$1k, no card debt	−1,164 (776)	−169 (196)	−0.1 (5.3)	−0.14 (0.12)	0.0 (5.3)
≤$1k, card debt	−2,612 (8,197)	−1,427 (2,297)	0.3 (13.8)	0.05 (0.80)	6.6 (13.1)
≤$1k, no card debt	805 (909)	−15 (364)	−2.9 (11.3)	0.07 (0.20)	−3.4 (11.1)
Single mothers					
>$1k, card debt	−1,395 (581)	−1,329 (353)	−24.2 (11.3)	−0.45 (0.18)	22.4 (11.7)
>$1k, no card debt	−232 (321)	−62 (202)	−0.6 (8.7)	−0.05 (0.11)	2.9 (8.0)
≤$1k, card debt	−401 (832)	−1,001 (499)	−15.9 (13.4)	−0.28 (0.25)	11.8 (14.1)
≤$1k, no card debt	991 (438)	463 (301)	1.7 (8.9)	0.27 (0.15)	0.1 (8.6)

Rows: groups by December 2020 liquid assets (≤ or >$1,000) and card debt. N and the paid-off or started shares: mothers / comparison women. Panel A: survey-weighted means; changes winsorized at the 1st and 99th percentiles, with replicate-weight standard errors; paid off (rows with card debt): share of women with card debt in December 2020 who had none in December 2021; started (rows without card debt): share of women without card debt in December 2020 who had some in December 2021. Panel B: coefficients on the mother indicator interacted with each group in equation (1), with group main effects and the controls of Table 4; "Main" repeats Table 5; "Share": difference in the paid-off or started share of Panel A. Single mothers: all 159.

For mothers with little cash, then, card debt did not detectably fall and bank balances showed no detectable increase relative to comparison women, and the data cannot say whether the credit changed their borrowing. Two readings are consistent with this, and annual balances cannot separate them. The advances may have been consumed as they arrived: cash-poor single mothers hold about $120 in the bank (married, $230), the Chase data show low-buffer families spending most of each advance in its first week, and the hardship studies show food insufficiency falling, so nothing was left to repay with. Or some mothers may have borrowed against the payment still to

come: balances are measured at the end of December, after the last advance and before the filing payment, and the relative increase in card debt is concentrated among cash-poor single mothers whose payment at filing was larger under the 2021 law than under old law ($1,059, SE 616, against −$381, SE 607, for those whose filing payment fell), although neither estimate is significant.

5.3 Constraint or impatience?

Low cash can reflect a temporary shock or a persistent pattern. A mother who has just lost income or paid a large bill holds little cash temporarily and rebuilds it when she can; a mother who discounts the future heavily, or who lives with persistently tight finances, holds little cash year after year (Aguiar et al., 2025). For the women in the 2020 SIPP panel we observe liquid assets in December 2019, 2020 and 2021. Among single mothers with $1,000 or less in December 2019, 81 percent (SE 8) still had $1,000 or less in December 2020, against 68 percent (SE 7) of comparison women; of single mothers low in both years, 84 percent (SE 8) were still low in December 2021, after six months of advances. Among married mothers, low cash is much less persistent: 47 percent (SE 9) of those low in 2019 were still low in 2020. These samples are small (36 single and 29 married mothers low in 2019), too small to compare responses by persistence, but the contrast is clear: low cash among single mothers is mostly a persistent state; among married mothers it is more often temporary.

Debt repayment helps separate constraint from impatience. Using an advance to repay a card balance at a 16.45 percent APR gives up consumption now for interest saved later, so whether a household repays depends on how much it needs the cash now as well as on its patience, and no single discount factor marks the line. A household that discounts the future steeply, or that needs the cash for current spending, has little reason to repay early. That single mothers with some cash repaid debt suggests that many of them valued the interest saved more than the cash in hand. Cash-poor mothers without card debt, by contrast, did not use the advances to build a buffer. One more piece of evidence points to tight cash flow among single mothers: in the CPS, single mothers' entry into work rose by 3.6 points (SE 1.5) in January and February 2022, the two months after the monthly payments stopped and before the filing payments arrived, and fell back in the filing season (Appendix Table 23).

5.4 Large bills and income drops arrive at uncertain times

Whether earlier payment smooths consumption depends on whether mothers face months in which cash runs short. Table 7 shows that such months are common. A married mother who pays for childcare pays $1,164 a month; a single mother pays $598, a larger share of her resources. The largest one-month drop in family income during 2021 averaged $3,504 for married and $1,008 for single mothers. These shocks are large next to the $230 and $120 median bank balances of cash-poor mothers.

Table 7: Childcare bills and income drops

	Married mothers		Single mothers	
	All	Liquid ≤$1k	All	Liquid ≤$1k
Childcare bill per month, payers ($)	1,164	1,071	598	692
Largest one-month income drop, 2021 ($)	3,504	1,707	1,008	700
Paid childcare, 2020 / 2021 (%)	24.5 / 24.6	9.9 / 17.1	28.7 / 30.3	20.4 / 20.7
Paid childcare started or stopped, 2020 to 2021 (%)	25.4	24.5	19.7	17.6
Starts and stops with a change in employment (%)	19	–	10	–

Model samples, weighted means. Largest drop: each mother's largest fall in monthly family income between consecutive months of 2021. Change in employment: employed in December 2020 but not December 2021 or the reverse. – : too few mothers.

Childcare spending is also hard to predict. Between 2020 and 2021, one in four married mothers and one in five single mothers started or stopped paying for childcare. Only one in five of these starts and stops coincided with a change in the married mother's employment, and one in ten for single mothers: children start school, a relative stops helping, a provider closes. We therefore treat childcare bills as cost shocks that arrive independently of the credit, with work held fixed.

5.5 What the evidence implies for a model

The evidence motivates a parsimonious specification rather than determining one. Table 8 pairs each fact with the feature of the model it motivates, and Sections 9.5 and 10 examine how the results change when those features change.

Table 8: From the reduced-form evidence to the model

Evidence (Sections 4 and 5)	Feature it motivates (Sections 6 to 10)
No detectable change in employment	Work held fixed; the credit valued through consumption
Spending and hardship follow payment dates; many mothers hold almost no cash	Cash on hand binds: in the main model mothers cannot borrow
Childcare bills and income drops arrive unpredictably	Income risk and a childcare-cost process, which give savers a precautionary reason to value cash early
Card debt fell among mothers with cash and debt; cash-poor mothers' bank balances showed no detectable relative increase	Two kinds of mothers: savers, who use early cash against expensive debt, and spenders, who consume what arrives
Low cash is persistent among single mothers	The spender type is a stable description, with a probability that depends on starting cash, not a saver who had a bad year
Four in ten married and half of single mothers carried card debt at the end of 2020	The borrowing extension: savers borrow at the card rate while cash earns nothing; spenders sit at their credit limit and pay interest on it

Two of these deserve comment. First, the card-debt split motivates allowing two kinds of mothers. Paying down a 16 percent balance with an advance is forward-looking behavior that requires a buffer

to live on meanwhile; a buffer-stock saver with little cash would instead put part of an advance toward a buffer, and the balances show cash-poor mothers who did not. One optimizing model can produce different responses when mothers differ in resources, debts, risks and needs, so the split does not establish two types. We adopt savers and spenders as a parsimonious description that the December balances fit better than savers alone or an impatient type (Section 7 and Appendix Table 26).

Second, the reduced form cannot say whether the credit changed borrowing among cash-poor mothers: the estimates rest on a few large balances, and comparison women differed from mothers in more than the credit. Nor can it separate the larger credit from its earlier payment. The model does both, because it solves each mother's problem under each calendar with everything else held fixed (Sections 9 and 10).

Both readings of the cash-poor pattern have a counterpart in the model: advances consumed as they arrive describe the spender, and borrowing against a payment due at filing is what a saver in the borrowing extension does when the credit is paid only at filing, a need the advances remove. Because the model is estimated on each mother's own December 2021 balance against its prediction for her, it needs no comparison group.

What the evidence does not settle is how the second kind of mother should be described. Mothers who keep nothing and mothers who keep a little but discount the future heavily both fit the direction of the evidence, and Section 9.5 shows that this is the choice the single-mother value is most sensitive to.

The facts cannot value earlier payment by themselves. Every eligible mother received the larger credit, the advances and the 2021 stimulus checks at the same time, so no comparison in the data isolates timing, and a spending response is not a welfare measure: an advance that pays the rent on time is worth more than one that moves a purchase by a few weeks. The model translates the timing of payments into dollars that are comparable across mothers.

6 A monthly model of saving under income and childcare risk

Mothers come in two kinds. *Savers* are buffer-stock savers (Deaton, 1991; Carroll, 1997): each month they decide how much to consume and how much cash to carry into the next month, facing random income and a childcare bill that can start or stop. *Spenders* consume all the cash they have each month, a description of the mothers whose balances stay near zero year after year (Section 5); whether a mother is a spender is a probability fitted to her balances, not an observed behavior. Work and family composition are fixed, so the credit affects mothers only through the budget. We solve the model separately for each mother, from her own resources, credits, childcare costs and starting cash.

6.1 The monthly problem

Time is monthly; $t = 0$ is July 2021. Let μ_i be mother i's average monthly resources in 2021 after federal income and payroll taxes, excluding the CTC and other refundable credits. At the start of month t she holds cash $a_t \geq 0$ and her paid-childcare status from the previous month, $d_{t-1} \in \{0, 1\}$. During the month:

1. Childcare status d_t is drawn. A new spell of paid care starts with probability s_j, and an

ongoing spell stops with probability q_j, where j indexes the age of the youngest child (0–4 or 5–12). When the youngest child is 13 or older, no new spells start.

2. Income $y_t = \mu_i \varepsilon_t$ arrives, with $\ln \varepsilon_t \sim N(-\sigma_g^2/2, \sigma_g^2)$ independent across months, where g indexes the mother's 2020 income group.
3. CTC receipts b_t^X under calendar X and other refundable credits r_t (in April) arrive, and the childcare fee $f_i d_t$ is paid.
4. She chooses consumption c_t and carries $a_{t+1} = a_t + y_t + b_t^X + r_t - f_i d_t - c_t \geq 0$ into the next month. Cash earns no interest and she cannot borrow.

If cash on hand falls below a consumption floor $\underline{c} = 0.1\,\mu_i$, as it can after a large fee and a low income draw, she consumes $\underline{c}$ and carries nothing; the floor stands in for public and private support not recorded in the data. The value function satisfies

$$V_t^X(a, d_-) = \sum_{d \in \{0,1\}} P_j(d \mid d_-)\, \mathbb{E}_\varepsilon \max_{0 \leq a' \leq \hat{a}(x)} \Big[u(\tilde{x} - a') + \delta\, V_{t+1}^X(a', d) \Big], \qquad x = a + \mu_i \varepsilon + b_t^X + r_t - f_i d, \tag{2}$$

where $\tilde{x} = \max(x, \underline{c})$ is cash after any top-up to the floor and $\hat{a}(x) = x\,\mathbf{1}[x \geq \underline{c}]$, so that a mother who needed the top-up carries nothing forward. Cash a, income and payments are in dollars; we solve the model with cash measured in months of resources, a/μ_i (Appendix B). Utility is $u(c) = \ln c$ and the monthly discount factor is $\delta = \beta^{1/12}$. We set the annual discount factor to $\beta = 0.96$, a standard value in buffer-stock models (Carroll, 1997), and report results for β from 0.90 to 0.99.

6.2 Spenders

A spender faces the same income, childcare bills, credits and consumption floor, but consumes all her cash on hand each month, $c_t = \max(x_t, \underline{c})$, and carries nothing forward. Spenders can be read as the limit of strong present bias (Laibson, 1997): a mother who barely weighs next month against this one spends whatever arrives, whatever her long-run patience. Their behavior therefore needs no discount factor. To value it we use the same β as for savers, the long-run criterion common in welfare analysis with present bias (O'Donoghue and Rabin, 1999; Bernheim and Rangel, 2009), so that savers and spenders differ in how they handle cash, not in how their wellbeing is counted. A spender's value of calendar X is

$$W_t^X(d_-) = \sum_d P_j(d \mid d_-)\, \mathbb{E}_\varepsilon \Big[u\big(\max(x, \underline{c})\big) + \delta\, W_{t+1}^X(d) \Big], \qquad x = \mu_i \varepsilon + b_t^X + r_t - f_i d, \tag{3}$$

plus her starting cash in July.

Valuing spenders at savers' patience is a normative choice, and the single-mother results depend on it. We make it for three reasons. First, the alternative needs a discount factor that the data cannot supply: a mother who keeps nothing does so at any discount factor low enough, so the balances identify that she keeps no buffer but not how impatient she is. Second, the policy question is how a payment calendar changes a mother's wellbeing over the year, which is what the long-run criterion measures; under it, six monthly payments do for a spender what she does not do for herself, spreading the credit over lean months. Third, the choice is easy to undo: Section 9.5 and Appendix Table 26 report values with an impatient type in place of spenders, counted both at that type's own discount factor and at savers', and the single-mother value ranges from about half the main value to well above it; Section 9.5 also values spenders' consumption at their own impatience, which raises their values.

6.3 Calendars, continuation and starting point

From July 2021 to June 2022 the receipts b_t^X follow the calendars in Table 1, and the April 2022 filing payment includes the mother's 2021 earned income and childcare credits under every calendar. From July 2022 every calendar continues with the same repeating year under 2022 law: monthly resources at their 2022-law level, the ordinary 2022-law credits (the $2,000 CTC, earned income and childcare credits) paid each April, and the same income and childcare risks. We solve this continuation as a stationary twelve-month cycle and use it as the terminal value for the first year.

The childcare fee f_i is the mother's own reported weekly payment converted to a monthly amount, from 2021 if she paid then and from 2020 otherwise. For mothers who paid in neither year, the fee a new spell would bring is $\phi_j \mu_i$, a share of her resources that depends on the youngest child's age. Fees are capped at 60 percent of monthly resources.

Each mother starts in July 2021 with

$$a_0 = \text{Bank}_i^{\text{Dec 2020}} + \kappa\, \text{EIP}_i^{\text{Jan–Jun 2021}}, \qquad d_{-1} = \mathbf{1}[\text{paid childcare in 2021}], \tag{4}$$

where κ is the share of the spring 2021 stimulus payment still held in July. We set $\kappa = 0$; in an earlier version that estimated patience, the estimate for married mothers was at this bound. A spender spends her starting cash in July. The SIPP records whether the family paid for childcare at some point during the reference year, not in which months, so the July 2021 status is set from the 2021 indicator: a mother who paid at some point in 2021 starts July paying, and the chain then moves her in and out of care. Section 9.5 replaces this with the probability that she pays in June 2021 given both annual indicators under the estimated chain, which averages 66 percent (survey-weighted) for single mothers who paid at some point in 2021. Two further features of the starting point are maintained assumptions rather than estimates: July cash is proxied by the December 2020 bank balance, without the spring 2021 stimulus or the tax year 2020 refundable credits received in spring 2021 (Section 9.5 adds part or all of each), and income from July on is drawn around the mother's 2021 monthly average, so the model does not use the January–June path of her finances.

6.4 The value of earlier payment

We express the value of each calendar as a grant on top of old-law payment, paid in twelve equal monthly parts from July 2021 to June 2022. For calendar X, the equivalent grant $m(X)$ solves

$$V_0^{O+m}(a_0, d_{-1}) = V_0^X(a_0, d_{-1}), \tag{5}$$

where $O + m$ adds $m/12$ to cash in each month of the first year, and V is replaced by W for spenders. The *amount* value is $A_i = m(F)$, the value of the expanded credit paid at filing. The *total* value is $T_i = m(S)$, the value of the expanded credit paid as in 2021. The *timing value* is their difference,

$$\tau_i = T_i - A_i = m(S) - m(F), \tag{6}$$

the value of receiving the same expanded credit half in advances rather than all at filing, measured on the same scale. When the text says what the 2021 calendar was worth, it means the timing value τ_i. Dividing by the added credit $\Delta B_i = B_i^N - B_i^O$ gives values per added credit dollar. We report four such measures, aggregated as ratios of weighted totals: the *filing-only value*, $100\, A_i/\Delta B_i$ cents per added dollar; the *total value*, $100\, T_i/\Delta B_i$; their difference, the *timing value per added dollar*,

$100\,\tau_i/\Delta B_i$; and the *cents lost* by paying only at filing, $100\,(1 - A_i/\Delta B_i)$. The total value can exceed 100 cents, because the grant measures the whole calendar and the advances also paid half of the old-law credit before filing. A saver who would hold positive cash in every month under every calendar values timing at zero: with no interest on cash, she can replicate the later payment by saving the advance and cannot gain by receiving it earlier. For savers, timing has value only if, under filing-only payment, they would or might run out of cash before April 2022. For spenders, any payment that arrives in a lump is consumed in that month, so spreading a payment over months has value of its own.

We spread the compensating grant over the year because a spender would consume a one-time grant in the month it arrived, so a lump-sum yardstick would mostly measure the cost of that lump rather than the value of the calendar. For savers the choice hardly matters: their timing value as an equivalent July 2021 grant, the measure in earlier versions of this paper, is within 3 percent of the spread measure on average (Table 14).

The model omits channels that could also make monthly payment valuable or costly: costs of administering monthly payments, self-control problems, uncertainty about the size of the refund (Caldwell et al., 2023), changes in tax withholding (Section 9.5 examines it), and effects on work. For savers the timing value is the value of earlier access to cash for a mother who saves optimally given her risks; for spenders it is the value of receiving the credit in smaller, earlier pieces.

7 Estimation

We set patience and estimate everything else by maximum likelihood from each mother's own records. No moments are targeted.

7.1 What is set and what is estimated

The annual discount factor is set to $\beta = 0.96$ for every mother. In principle December balances carry information about patience, and an earlier version of this paper estimated β from them. The difficulty is the mothers who end 2021 with almost no cash: any discount factor low enough makes a saver keep nothing, so balances can show that a mother keeps no buffer but not how impatient she is. A common practice in buffer-stock work is to set β and show how results depend on it (e.g., Carroll, 1997; Kaplan and Violante, 2014); we do so, and replace a single low discount factor with a second kind of mother, the spender, whose share the balances do identify. Section 9 shows that between $\beta = 0.90$ and $\beta = 0.99$ the single-mother values change by less than 10 percent and the married average ranges from \$173 to \$247, because much of the value of earlier payment comes from the mothers who keep no cash rather than from how savers discount. The likelihood of the December balances rises with savers' discount factor over this range, by 8.1 log points from 0.96 to 0.99 (Appendix Table 25), so the data favor more patience, not less, than we assume; at $\beta = 0.99$ the average values are 4 percent (single) and 10 percent (married) below those at 0.96. Utility is logarithmic (relative risk aversion of one; Section 9.5 reports 2 and 3), cash earns no interest, the consumption floor is 10 percent of resources and the fee cap is 60 percent.

The estimated parameters are income risk, from monthly income records, the childcare process, from the annual childcare indicators, and fee shares, from reported weekly payments, together θ_1; and, from December 2021 balances, the probability that a mother is a spender and the reporting error, θ_2. The log likelihood of mother i is $\ell_i = \ell_{1i}(\theta_1) + \ell_{2i}(\theta_1, \theta_2)$.

7.2 Income, childcare and fees

Income. The model's income process implies that a mother's log monthly family income deviates from its own mean by independent normal shocks. After removing common month effects, the deviations of each mother's twelve 2021 monthly incomes contribute $-\frac{11}{2}\ln\sigma_g^2 - \frac{Q_i}{2\sigma_g^2}$, where Q_i is the sum of squared deviations. For married mothers the three income groups are 2020 family income below \$50,000, \$50,000–100,000 and above \$100,000; for single mothers, whose incomes are lower, below \$25,000, \$25,000–50,000 and above \$50,000.

Childcare. The SIPP records whether the family paid for childcare at some point during each reference year, not in which months. The likelihood therefore aggregates the monthly chain to that observation: with the December 2019 status drawn from the chain's stationary distribution, we compute the probability that a mother pays in at least one month of 2020 and in at least one month of 2021 by tracking, month by month, her status and whether she has paid yet that year. The two annual indicators contribute $\ln \Pr(D_{i,2020}, D_{i,2021} \mid s_j, q_j)$, where $D_{i,y}$ is the indicator for year y. Two indicators identify the two rates through the share of mothers who pay in a given year and the persistence of paying across years; spells that start and stop within a year are counted, since the indicator records them. Among the 147 single mothers only 41 have a youngest child under five, so for single mothers we pool the start and stop rates across the two age groups; fee shares remain age-specific.

Fees. Among 2021 payers, the log of the monthly fee relative to resources, $\ln(f_i/\mu_i)$, is normal with mean $\ln\phi_j$.

7.3 December 2021 balances: savers and spenders

A saver's model delivers a predictive distribution G_i of her bank balance in December 2021 under the 2021 calendar, starting from her December 2020 balance; a spender ends every month with nothing. We give every mother the statutory advance schedule. Reported advances are lower: \$2,002 for married and \$1,818 for single mothers against scheduled amounts of \$2,819 and \$2,743, and 29 and 25 percent of mothers report no advance at all. Whether this reflects non-receipt or under-reporting is unclear, so Section 9.5 re-estimates the model with each mother's reported monthly receipts.

The observed balance b_i is reported with error on the inverse hyperbolic sine scale, $z_i = \operatorname{asinh}(b_i/1000) = \max\left(0, \operatorname{asinh}(\tilde{b}_i/1000) + e_i\right)$, where $\tilde{b}_i$ is her true balance, so that reports are censored at zero, as balances are. The error e_i is normal with standard deviation σ_e (by marital status) with probability $1-\omega$ and, with probability ω, a wide normal with standard deviation 2.2, so that a few large balance changes are not read as behavior. The outlier component is a device for robust estimation rather than a model of how balances are reported (Section 8). A saver's contribution is $\mathcal{L}_i^{\text{sv}} = \int h(z_i \mid \tilde{b})\, dG_i(\tilde{b}; \theta_1)$ and a spender's is $\mathcal{L}_i^{\text{sp}} = h(z_i \mid 0)$, where h is the censored-normal density. The probability of being a spender is

$$p_i = \Lambda\left(\alpha_{\text{mar}(i)} + \gamma_b \operatorname{asinh}(\text{Bank}_i^{\text{Dec 2020}}/1000)\right), \tag{7}$$

where Λ is the logistic function, with an intercept for each marital status, α_{married} and α_{single}, and a common slope γ_b on starting cash. Letting the type depend on the initial condition is the standard treatment of the initial-conditions problem in dynamic models with unobserved heterogeneity (Wooldridge, 2005): a mother who keeps nothing in December 2021 should usually have kept little

a year earlier. Then

$$\ell_{2i} = \ln\left[p_i \mathcal{L}_i^{\text{sp}} + (1 - p_i)\mathcal{L}_i^{\text{sv}}\right]. \tag{8}$$

The balances identify the spender probability because the two kinds of mothers predict different things for a mother who starts with little cash: a saver facing income and childcare risk builds a buffer out of the advances, while a spender does not. The error term absorbs misreported balances and household finances outside the model.

7.4 Procedure and inference

We estimate θ_1 from ℓ_1 and then θ_2 by maximizing $\sum_i w_i \ell_{2i}$, with survey weights w_i, given $\hat{\theta}_1$, pooling married and single mothers, whose spender intercepts and reporting-error variances differ but who share the slope and the outlier probability. Standard errors are design-based: for each of the 240 replicate weights we re-estimate θ_1 and re-fit θ_2, so the standard errors include first-step uncertainty. In the replicate fits each mother's saver likelihood and welfare are linearized in θ_1, using model solutions at perturbed values of each first-step parameter (Appendix B).

7.5 Estimates

Table 9: Parameter estimates

	Married mothers (593)		Single mothers (147)	
Income risk σ_g, by 2020 family income				
Lowest group	0.293 (0.037)	[<$50k]	0.347 (0.062)	[<$25k]
Middle group	0.189 (0.021)	[$50–100k]	0.337 (0.074)	[$25–50k]
Highest group	0.220 (0.023)	[≥$100k]	0.291 (0.084)	[≥$50k]
Childcare, monthly rates (%)				
Start, youngest 0–4	1.93 (0.30)		1.44 (0.34)	pooled
Start, youngest 5–12	1.43 (0.22)			
Stop, youngest 0–4	8.76 (2.48)		5.03 (2.33)	pooled
Stop, youngest 5–12	21.44 (8.81)			
Fee share of monthly resources ϕ_j (%)				
Youngest 0–4	9.94 (0.84)		21.46 (4.48)	
Youngest 5–12	4.93 (0.67)		12.08 (4.28)	
Savers and spenders (December 2021 balances)				
Spender share (%)	2.3 (1.5)		31.1 (5.6)	
Intercept α_{married}, α_{single}	−0.08 (0.94)		2.74 (1.33)	
Slope on asinh(December 2020 bank/1000), γ_b	−6.98 (6.78), common			
Reporting-error SD σ_e (asinh scale)	1.009 (0.064)		0.620 (0.192)	
Outlier probability ω	18.4% (5.2), common			
Set				
Annual discount factor β	0.96			

Maximum likelihood. Design-based standard errors from 240 replicate weights, re-estimating the first step in each replicate. Re-fitting from 16 random starting values raises the log likelihood by at most 0.03. Spender share: weighted mean of the prior probability p_i. Log utility; stimulus carry-over share $\kappa = 0$.

Table 9 reports the estimates. Married mothers' monthly income is volatile, with a standard deviation of log shocks of 0.29 in the lowest income group and about 0.2 in the others. Paid childcare spells start at 1.4 to 1.9 percent a month and stop at 9 to 21 percent, faster for school-age

children: a spell lasts about a year on average for a young child and five months for a school-age child, and a new spell costs 10 percent of monthly resources for a young child and 5 percent for a school-age child. Single mothers face more income risk in every income group; their spells are rarer but longer (1.4 percent a month to start, 5 percent to stop), and their childcare fees are more than twice as large relative to their resources, 21 percent for a young child and 12 percent for a school-age child.

The fitted model assigns a large spender probability to single mothers and a small one to married mothers: on average 31 percent (SE 5.6) against 2.3 percent (SE 1.5). Spenders are concentrated among mothers who started 2021 with almost no cash. The slope on starting cash is steep, so that a single mother with an empty account in December 2020 has a 94 percent prior probability of being a spender and one with $1,000 has a 3 percent probability; the magnitude of the slope is imprecise, but the likelihood ratio for including it is 31.4 (1 degree of freedom). These are probabilities fitted to annual balances; they say that the balances of these mothers look like those of someone who keeps nothing, not that any mother is observed spending.

Table 10: Savers and spenders: model comparison and posterior shares

Model comparison (740 mothers)	Log likelihood	Parameters	BIC
Savers only	−1,253.5	3	2,526.9
Savers and spenders, one spender share	−1,245.3	4	2,517.0
shares by marital status	−1,233.6	5	2,500.2
shares by marital status and December 2020 cash (main)	−1,217.9	6	2,475.4
Posterior spender share by group (%)	Married	Single	
All	2.3 (1.6)	31.2 (5.6)	
Liquid assets ≤$1k	20.9 (13.7)	64.5 (9.4)	
Liquid assets $1k–25k	0.0 (0.1)	0.6 (2.1)	
Lower income (married <$50k; single <$25k)	9.6 (6.5)	53.3 (10.9)	
Card debt in December 2020	0.8 (0.9)	27.8 (6.6)	
No card debt in December 2020	3.2 (2.1)	35.5 (7.6)	

Likelihood of the December 2021 bank balance, with income, childcare and fee parameters at each sample's first-step estimates and $\beta = 0.96$. All specifications include the censored reporting error and the outlier component. Posterior shares: weighted means of each mother's posterior probability of being a spender given her December 2021 balance. Design-based standard errors.

Table 10 compares specifications. Allowing for spenders improves the fit (likelihood ratio 16.4 for one common share), letting the share differ by marital status improves it further (23.5), and letting it depend on starting cash improves it again (31.4); the main specification has the lowest BIC. These posterior probabilities are model-based classifications, not observed behavior. Among single mothers with $1,000 or less in liquid assets, the posterior spender probability averages 65 percent, and among those with income below $25,000, 53 percent. Among married mothers with $1,000 or less it averages 21 percent, with a wide standard error. Single mothers who carried card debt are assigned lower spender probabilities than those who did not (27.8 against 35.5 percent), in line with the card-debt evidence of Section 5; among cash-poor single mothers the gap (59 against 72 percent) comes mostly from their starting cash, which already sets the prior at 61 against 69 percent. Above $1,000 of liquid assets almost no mother is classified as a spender.

8 How well does the model fit?

The likelihood uses each mother's December 2021 balance. No summary of the balance distribution is targeted, but the type probabilities are estimated from these same balances, so the comparisons below are in-sample checks rather than independent validation. Table 11 compares the model's predicted December 2021 balances with the data, with a model of savers only, and with the naive prediction that each mother's balance stays at its December 2020 level. The model columns exclude reporting error; the column "Model, reported" adds the estimated reporting error. Figure 4 plots the full distributions.

Table 11: December 2021 bank balances: data and model

		Observed	Model	Model, reported	Savers only	December 2020
Married mothers						
All (593)	Mean ($)	59,049	45,162	154,685	45,239	51,886
	Median ($)	16,342	12,987	13,500	12,987	15,000
	Share ≤$1,000 (%)	10.0	5.0	10.7	3.2	11.3
Liquid ≤$1k (71)	Mean ($)	6,558	2,625	9,513	3,319	332
	Median ($)	800	2,029	1,854	2,655	230
	Share ≤$1,000 (%)	54.7	32.1	38.4	15.2	100
Single mothers						
All (147)	Mean ($)	9,295	7,042	22,002	7,702	7,137
	Median ($)	1,300	2,259	2,130	3,037	1,100
	Share ≤$1,000 (%)	45.0	37.1	38.3	13.6	48.1
Liquid ≤$1k (72)	Mean ($)	2,247	1,023	4,019	2,395	300
	Median ($)	300	0	386	1,922	120
	Share ≤$1,000 (%)	66.7	71.5	66.1	22.6	100

Model: each mother's predictive distribution of her December 2021 bank balance under the 2021 calendar, a mixture of her saver prediction and a spender's zero balance with her prior spender probability, aggregated with survey weights. Model, reported: the same with the estimated reporting error (normal and outlier components on the inverse hyperbolic sine scale, censored at zero). Savers only: the saver predictions alone. December 2020: each mother's own balance a year earlier.

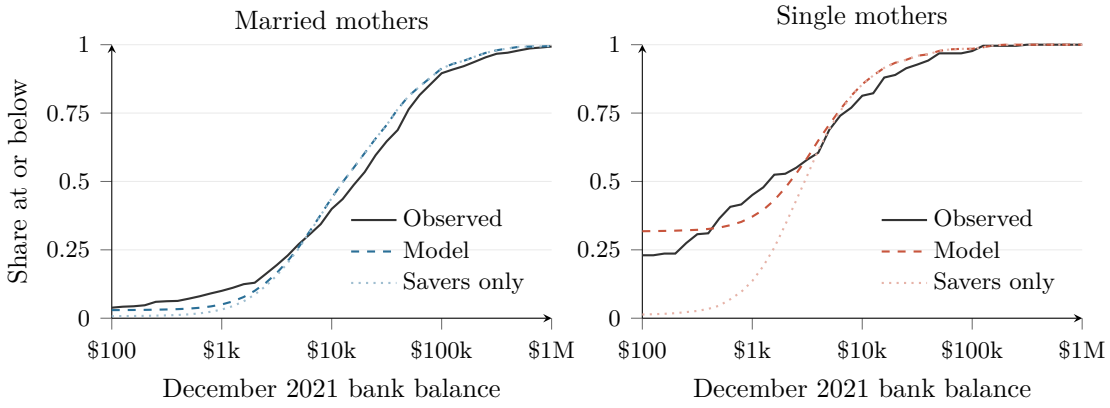


Figure 4: Cumulative distribution of December 2021 bank balances: share of mothers with a balance at or below each amount (log scale), all mothers in each model sample, survey-weighted. Model: each mother's saver prediction mixed with a spender's zero balance, using her prior spender probability, without reporting error. Savers only: every mother predicted as a saver. A model of spenders only would place every mother at zero.

The spenders are what let the model reproduce the mass of mothers near zero. A model of savers alone predicts that 14 percent of single mothers hold $1,000 or less at the end of 2021, against 45 percent in the data: facing their income and childcare risks, savers would build buffers out of the advances. With spenders the model predicts 37 percent, and among single mothers who started with $1,000 or less it predicts 72 percent against 67 percent observed. For married mothers the improvement is smaller, because the spender group is small: among those who started with $1,000 or less, 55 percent still held $1,000 or less a year later, and the model predicts 32 percent (15 percent with savers only). The model predicts a median balance of $2,300 for single mothers against $1,300 observed.

Two cautions apply. First, the naive prediction that balances stay at their December 2020 level is closer than the model to the observed medians in both full samples and to the share at or below $1,000 among single mothers (for married mothers, the model with reporting error comes closer to that share: 10.7 percent against 10.0 observed and 11.3 naive). The naive prediction is not a model of how balances respond to the credit, but it shows that much of the cross-section of balances is persistence that the model reproduces only in part. Second, the reporting error matters for means. With the estimated error included, the model comes close to the share at or below $1,000 for all married mothers (10.7 against 10.0 percent) and for cash-poor single mothers (66.1 against 66.7 percent), and less close in the other groups, but the implied reported means are far above the observed ones ($154,685 against $59,049 for married mothers), because an error on the inverse hyperbolic sine scale is roughly proportional in dollars and the outlier component, which covers 18 percent of reports, is wide. We therefore treat the outlier component as a device for robust estimation rather than as a description of how balances are reported, and the fit of balances as a check that the model is not grossly wrong about where mothers end the year, not as a validation of the behavioral classification. Without the outlier component the fit is worse (log likelihood −1,236.2 against −1,217.9); the average welfare values change little, although the values for cash-poor mothers fall by 12 to 17 percent (Table 15).

A further check uses card debt. Section 10 lets savers borrow at credit-card rates. For married mothers, that model predicts that the 2021 calendar lowered December 2021 card debt by $233 to $459 per mother relative to old-law payment, depending on the credit limit, or 8 to 16 cents per

scheduled advance dollar. The regression in Table 4 implies a decline of $1,909, about two thirds of the scheduled advances, so the model's decline is between an eighth and a quarter of the estimate. The model understates the decline because it nets cash against card debt: most married mothers with card balances also held more cash than they owed, so the model treats them as savers with positive cash. Section 10 extends this check to single mothers and to groups by starting cash and card debt (Table 19). How the remaining gaps bear on the welfare results depends on what is missing from the model, so we do not sign their effect.

A third check looks a year beyond the data used in estimation. The 2023 SIPP reports December 2022 bank balances for 424 of the married and 96 of the single mothers, after the spring 2022 filing payment, part of which a saver would have kept. Holding every estimate fixed, the model forecasts that 60 percent of cash-poor single mothers report $1,000 or less in the bank in December 2022, against 55 percent observed; with savers only and the same reporting error, it would forecast 16 percent. For cash-poor married mothers the forecast is 36 percent, against 48 percent observed. The naive forecast that each mother's balance stays at its December 2021 level gives 63 and 52 percent, a little further off for single mothers and closer for married mothers: low balances that persist through the filing payment are what spenders imply and savers do not (Appendix D.7). The check therefore speaks to the levels of balances and to how cash-poor mothers are described, not to the value of earlier payment.

9 The value of earlier payment

9.1 Main results

Table 12 reports, for each group, the average added credit, the timing value τ, the value of the expanded credit paid only at filing per added credit dollar, the timing value as a share of the advanced half of the credit, and the posterior share of spenders. Each mother's values average her saver and spender values with her posterior probabilities. All values are equivalent grants spread over July 2021–June 2022, and the standard errors include parameter uncertainty. The calendars are the statutory ones: each mother receives the advances and filing payment the law scheduled for her. Mothers report receiving less. Single mothers reported 66 percent of their scheduled advances, and 25 percent reported none; married mothers reported 71 percent, and 29 percent none. If the gap is under-reporting, the statutory values apply; if some mothers did not receive advances, the values overstate what the 2021 calendar delivered to them. Valued at the advances they report receiving, with the rest of the credit at filing, the 2021 calendar was worth $660 (sampling SE 93) per single mother and $139 (sampling SE 29) per married mother, 67 and 73 percent of the statutory values, close to the share of scheduled advances reported (66 and 71 percent; Table 15).

Table 12: Value of the 2021 calendar relative to paying the same credit at filing

	N	Added credit ($)	Timing value τ ($)	Filing only: cents per added $	Timing, % of advance	Spenders (%)
Married mothers						
All	593	1,958	191 (37)	92.3 (1.8)	6.8	2
Liquid assets ≤$1k	71	3,063	993 (294)	71.6 (8.9)	29.2	21
Liquid assets $1k–25k	255	2,400	185 (20)	94.6 (0.5)	5.8	0
Liquid assets ≥$25k	267	1,260	6 (2)	99.9 (0.1)	0.2	0
Income <$50k	94	3,571	763 (182)	80.8 (5.0)	20.5	10
Income $50k–100k	146	2,333	136 (26)	96.1 (0.8)	4.4	2
Income ≥$100k	353	1,364	61 (10)	97.6 (0.4)	2.4	0
Paid childcare in 2021	156	1,799	225 (57)	92.0 (2.3)	8.0	1
No paid childcare	437	2,009	181 (39)	92.3 (2.0)	6.4	3
Single mothers						
All	147	2,758	987 (127)	60.1 (4.5)	36.0	31
Liquid assets ≤$1k	72	3,375	1,820 (203)	37.2 (5.0)	61.0	65
Liquid assets $1k–25k	59	2,306	269 (74)	91.4 (2.6)	10.4	1
Liquid assets ≥$25k	16	1,729	28 (19)	98.6 (1.0)	1.2	0
Income <$25k	51	3,853	1,685 (233)	44.1 (6.2)	57.4	53
Income $25k–50k	50	2,654	1,074 (213)	60.6 (7.6)	38.2	37
Income ≥$50k	46	1,802	188 (62)	93.1 (2.1)	7.6	2
Paid childcare in 2021	32	2,122	796 (160)	66.0 (7.5)	33.3	21
No paid childcare	115	3,035	1,070 (157)	58.3 (5.0)	36.9	36

Equivalent grants paid in equal monthly parts from July 2021 to June 2022, at $\beta = 0.96$; weighted means (timing) and ratios of weighted totals (cents per added dollar). Timing value $\tau = m(S) - m(F)$. Filing only: $m(F)$ per dollar of added credit. Advance: half of the expanded credit. Spenders: weighted mean posterior probability. Standard errors in parentheses include parameter uncertainty. Liquid assets in December 2020; income: 2020 family income.

For the average married mother, the 2021 calendar was worth $191 more than receiving the same expanded credit at filing, about 7 percent of the advanced half. Paid only at filing, each added credit dollar would have been worth 92 cents to her; with advances it was worth 102 cents, slightly more than a dollar because the advances also moved half of the old-law credit ahead of filing. For the average single mother the timing value was $987, more than a third of the advanced half. Paid only at filing, the expanded credit would have been worth 60 cents per added dollar, and even with the advances it was worth 96 cents, because the half paid at filing still arrived as a lump.

The value of earlier payment is concentrated among mothers with little cash. Married mothers with $1,000 or less in liquid assets valued the 2021 calendar at $993, or 29 percent of their advance; married mothers with $25,000 or more valued it at $6. Among single mothers the corresponding values are $1,820 and $28. Figure 5 shows the cents per added dollar lost by paying only at filing, by liquid assets. In each sample the loss falls steeply with cash on hand, and among mothers with little cash the single-mother loss is more than twice the married loss.

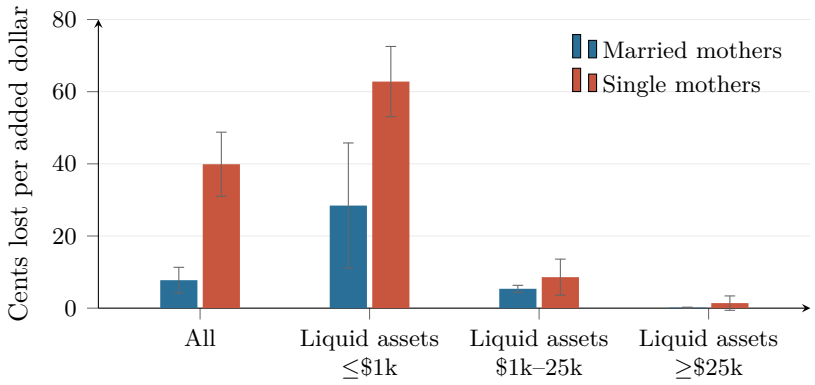


Figure 5: Cents lost per added credit dollar if the expanded 2021 credit had been paid entirely at filing, $100\,(1 - m(F)/\Delta B)$, by December 2020 liquid assets. Bars: 95% confidence intervals including parameter uncertainty.

9.2 Cash or income?

Low cash and low income go together, so Table 13 cross-classifies mothers by both. Within each income group, mothers with little cash gain much more from earlier payment. Married mothers with income of $100,000 or more but $1,000 or less in the bank would have lost 12 cents per added dollar from filing-only payment, against nothing for low-income married mothers with $25,000 or more in the bank. Among single mothers, those with $25,000 or more of income but $1,000 or less in liquid assets would have lost 54 cents per dollar, far more than low-income single mothers with more cash (14 cents). Income also matters, because lower-income mothers gain a larger credit and the lowest income groups face the most income risk. Much of the cash gradient within income groups comes from the spender classification, whose probability falls with starting cash. Valued as savers, low-income single mothers with $1,000 or less in liquid assets would lose 17 cents per added dollar from filing-only payment, against 13 cents for those with more; low-income married mothers with $1,000 or less in the bank would lose 17 cents as savers, against nothing for those with $25,000 or more. These cells are descriptive: cash is not randomly assigned.

Table 13: Timing value by income and cash

Income	Cash, December 2020	N	Timing value ($)	Filing only: cents per added $	Spenders (%)
Married mothers (bank balance)					
<$50k	≤$1k	41	1,435 (390)	67.9 (9.5)	24
≥$100k	≤$1k	15	396 (113)	87.7 (3.6)	13
<$50k	≥$25k	6	0 (0)	100.0 (0.0)	0
≥$100k	≥$25k	188	0 (0)	100.0 (0.0)	0
Single mothers (liquid assets)					
<$25k	≤$1k	39	2,256 (246)	29.9 (4.9)	79
≥$25k	≤$1k	33	1,488 (278)	46.1 (8.2)	54
<$25k	>$1k	12	513 (202)	86.3 (5.8)	1
≥$25k	>$1k	63	151 (44)	94.7 (1.4)	0

As in Table 12. Married cells use the December 2020 household bank balance; single-mother cells use liquid assets and coarser groups because of sample size. Small cells are imprecise.

9.3 Savers and spenders

Table 14 separates the two kinds of mothers. A saver values the 2021 calendar because she might run out of cash before filing; a spender values it because she consumes whatever arrives, so a credit paid in six monthly pieces raises consumption in six lean months instead of one month of plenty. Spenders' values are large: $1,132 for the average married mother and $1,824 for the average single mother, if each were a spender. Savers' values are much smaller, $152 and $476, and fall steeply with cash.

Table 14: Timing value for savers and spenders

	Spenders (%)	Savers ($)	Spenders ($)	Weighted ($)	Savers, July grant ($)
Married mothers					
All	2.3	152 (15)	1,132 (51)	191 (37)	149
Liquid assets ≤$1k	20.9	632 (72)	2,134 (221)	993 (294)	617
Liquid assets $1k–25k	0.0	185 (20)	1,457 (83)	185 (20)	182
Liquid assets ≥$25k	0.0	6 (2)	574 (40)	6 (2)	6
Single mothers					
All	31.2	476 (54)	1,824 (118)	987 (127)	467
Liquid assets ≤$1k	64.5	761 (80)	2,247 (164)	1,820 (203)	747
Liquid assets $1k–25k	0.6	260 (64)	1,560 (172)	269 (74)	255
Liquid assets ≥$25k	0.0	28 (19)	935 (218)	28 (19)	28

Timing value τ in dollars. Savers and Spenders: every mother in the group valued as that type. Weighted: posterior-weighted, as in Table 12; it is not the probability-weighted mix of the two columns (for single mothers $0.31 \times 1{,}824 + 0.69 \times 476 = \$894$, not $987), because each mother's spender probability is largest where the gap between her spender and saver values is largest. Last column: savers' timing value measured as an equivalent July 2021 grant. Standard errors include parameter uncertainty.

The spender value is a statement about the timing of consumption, not an estimate of preferences. It is the value, to a mother who consumes cash when it arrives, of receiving the advanced half of the credit in six monthly payments instead of with the filing payment, with her consumption counted at the savers' discount factor. Because she cannot spread a lump herself, the value is large relative to the credit: valued as spenders, single mothers would value the 2021 calendar at 66 percent of their added credit (67 percent for those with $1,000 or less in liquid assets), and married mothers with $1,000 or less at 70 percent. For 15 percent of single mothers and 16 percent of married mothers the spender value exceeds the added credit, mostly because the advances also moved half of the old-law credit ahead of filing; for 4 percent of single mothers it exceeds everything that was advanced, because a lump consumed in one month is worth little at the margin to a mother with low resources. The balances identify which mothers keep no cash, not why; the welfare rule is an assumption, and Section 9.5 shows that counting spenders' consumption with their own impatience would raise their values.

9.4 Why single mothers value timing more

Two differences make earlier payment worth more to single mothers. As savers, single mothers value the 2021 calendar at 3.1 times the married value ($476 against $152), because they hold smaller buffers, face more income risk and larger childcare fees relative to their resources, and gain

larger credits. And far more of them are classified as spenders. Much of the gap in spender shares reflects single mothers' lower starting cash: with the married intercept, so that only starting cash and balances shape their type, 19 percent of single mothers would be spenders, and their average timing value would be $822 (sampling SE 151) instead of $987.

9.5 How much do the assumptions matter?

Tables 15 and 16 collect the main alternatives.

Measurement. The reported-receipt row changes only the estimation of the type probabilities: each mother's saver prediction uses her reported July–December advances, so a mother who reports none is predicted to end 2021 with less cash as a saver, which lowers the spender share (29 percent of single mothers, 1.9 percent of married). Welfare still compares the statutory calendars, so this row shows the stability of the conditional estimates, not the value of the payments mothers report receiving: the values change by 3 to 6 percent. Precision falls more than the point estimates move: the sampling standard error of the cash-poor married value rises from $294 to $337, and that of the cash-poor single-mother value from $199 to $215. The statutory schedule fits the balances better (−1,217.9 against −1,225.6), which is consistent with under-reporting of receipts but could also reflect other misspecification of resources or behavior. The next row values the calendars mothers report, each mother's reported July–December advances with the rest at filing, at the main estimates: the values fall by 27 to 34 percent, about the shortfall in reported receipts (29 percent for married and 34 percent for single mothers); with the type probabilities estimated from reported receipts they are $136 and $645. Dropping the outlier component of the reporting error changes the averages by less than 4 percent and lowers the values for cash-poor mothers by 12 to 17 percent.

Starting conditions. Adding half of the spring 2021 stimulus to July cash lowers the averages by 6 to 11 percent and raises the married cash-poor value, because more married mothers who still held stimulus cash are then classified as spenders; adding all of it moves the values by up to about 20 percent and fits the balances markedly worse. Setting July childcare status from the chain rather than the annual indicator changes the values by at most 6 percent and fits slightly better. Adding the refundable credits that came with tax year 2020 returns in spring 2021 raises the averages instead. We compute them from 2020 earnings as the EITC and the refundable child credit, without refunds of over-withholding, which the SIPP does not record, or the option to use 2019 earnings; they averaged $2,901 for single and $960 for married mothers. Holding half or all of these credits in July fits the balances much worse (−1,233.8 and −1,247.1): savers then start with more cash and value timing less, but more cash-poor mothers are classified as spenders, and the averages rise by up to 12 percent (single) and 15 percent (married).

Work. Holding work fixed matters most for mothers who entered work during 2021, about one in ten single and one in thirteen married mothers (not employed in December 2020, working later in the year). Valued without their earnings from their first month of work, with taxes and credits recomputed and the type probabilities as estimated from the balances they actually reached, these mothers have less income and larger added credits, but the averages change by at most 3 percent: $962 for single and $193 for married mothers, and $963 and $193 if only entrants after June 2021 are treated this way. Without their earnings, half of the single-mother entrants would have less than $500 a month, the sample floor, at which we set their resources. The exercise lowers resources under every calendar alike, so it asks how much the values depend on entrants' earnings, not whether work responded to the timing of the payments, as the rise in single mothers' entry in early 2022

Table 15: Timing value under alternative specifications

		Married τ ($)		Single τ ($)	
	Log L	All	Liq. ≤$1k	All	Liq. ≤$1k
Main specification	−1,217.9	191	993	987	1,820
Measurement					
Reported advance receipts in the likelihood	−1,225.6	185	935	956	1,759
Reported advance receipts valued, rest at filing	–	139	686	660	1,194
No outlier component in the reporting error	−1,236.2	194	826	1,022	1,593
Starting conditions					
Half of the spring 2021 stimulus still held in July	−1,219.7	170	1,050	925	1,715
All of the spring 2021 stimulus still held in July	−1,231.8	178	1,196	958	1,724
Half of the 2020 refundable credits still held in July	−1,233.8	202	1,174	1,017	1,770
All of the 2020 refundable credits still held in July	−1,247.1	219	1,246	1,106	1,783
July childcare status from the chain (posterior)	−1,214.9	190	1,047	967	1,785
Work					
2021 work entrants without their new earnings	–	193	1,003	962	1,781
July–December entrants without their new earnings	–	193	999	963	1,787
Patience of savers					
$\beta = 0.90$	−1,243.6	247	1,014	1,073	1,901
$\beta = 0.99$	−1,209.7	173	1,072	946	1,773
Risk aversion					
Relative risk aversion 2	−1,211.0	311	1,614	1,202	2,094
Relative risk aversion 3	−1,209.4	408	1,990	1,274	2,084
Cash-poor mothers					
Savers only	−1,253.5	152	632	476	761
Impatient type ($\beta = 0.70$), own discount factor	−1,235.0	205	1,120	819	1,476
Impatient type ($\beta = 0.70$), valued at 0.96	−1,235.0	157	671	514	840
Impatient type ($\beta = 0.90$), valued at 0.96	−1,246.5	153	641	483	775
Impatient type ($\beta = 0.10$), valued at 0.96	−1,219.8	164	740	668	1,158

Timing value τ in dollars (equivalent spread grants). Log L: likelihood of the December 2021 balances of the 740 mothers, re-estimating the type probabilities in each row; –: main estimates. Reported receipts in the likelihood: each mother's reported July–December 2021 advances, with the rest of the credit at filing, in the likelihood only; welfare compares the statutory calendars. Reported receipts valued: the same calendar valued against filing-only payment. Stimulus: the mother's reported January–June 2021 economic impact payments added to July cash. 2020 refundable credits: the EITC and refundable child credit for tax year 2020, computed from 2020 earnings (refunds of over-withholding are not observed). Posterior July status: the probability of paying in June 2021 given both annual indicators under the estimated chain. Work entrants: mothers not employed in December 2020 (or June 2021) who worked later in 2021, valued without their earnings from their first month of work under every calendar, with taxes and credits recomputed and resources floored at $500 a month; type probabilities as estimated with actual earnings. Risk aversion: CRRA utility in savers' choices and in valuing both types. Re-estimating the main specification the same way gives log L −1,217.85 and values within 2 percent of the main row. Impatient type: Appendix Table 26. Further alternatives: Table 16.

Table 16: Timing value under other welfare rules, withholding, permanent payment and card credit

	Married τ ($)		Single τ ($)	
	All	Liq. ≤$1k	All	Liq. ≤$1k
Main specification	191	993	987	1,820
Valuing spenders' consumption				
Own discount factor, 0.70	201	1,080	1,082	2,017
Own discount factor, 0.50	211	1,172	1,181	2,222
Present bias, 0.70	195	1,030	1,027	1,903
Present bias, 0.50	200	1,075	1,076	2,005
Withholding and permanent payment				
Lower withholding under both calendars	112	711	868	1,652
2021 design made permanent, per year: switching	153	614	813	1,586
long-run states compared	58	518	748	1,549
Whole credit paid monthly, per year: switching	214	997	1,433	2,840
long-run states compared	98	872	1,347	2,792
Credit				
Card credit, three extra months of resources	119	636	797	1,557

Timing value τ in dollars, at the main estimates (type probabilities not re-estimated). Valuing spenders: spenders' consumption valued at an annual discount factor of 0.70 or 0.50, or by the July 2021 self with present bias of 0.70 or 0.50 on top of the main discount factor. Withholding: under both calendars, mothers lower July–December withholding by up to their July–December income tax before credits to receive the part of the credit not yet paid. Permanent: the expanded credit paid every year under the 2021 design or in twelve equal monthly payments, against all at filing; equivalent permanent monthly grant times 12. Switching: for a mother who starts from the long-run cash distribution under filing-only payment and switches, so the value includes the one-time release of the buffer she no longer needs to hold. Long-run states compared: each calendar at its own long-run distribution. Card credit: Section 10.

suggests some did (Section 5).

Patience of savers. Figure 6 recomputes the timing values for β between 0.90 and 0.99, re-estimating the spender model each time (Appendix Table 25). The average married value ranges from \$247 to \$173, and the average single-mother value from \$1,073 to \$946. More patient savers keep larger buffers and value timing less, while the estimated spender shares rise, from 1.3 to 3.4 percent among married mothers and from 30.2 to 32.2 percent among single mothers. For mothers with \$1,000 or less, the values stay within 10 percent of the main values.

Risk aversion. With relative risk aversion of 2 or 3, in savers' choices and in valuing both types, savers keep larger precautionary buffers but weigh a shortfall more heavily, spenders' lean months cost more, and the values rise: to \$1,202 and \$1,274 for single and \$311 and \$408 for married mothers. The balances fit better with more risk aversion ($-1{,}211.0$ and $-1{,}209.4$), as they do with more patience, which moves the values the other way; we do not estimate the two jointly.

Cash-poor mothers. How the mothers who keep little cash are described matters most. If they are savers like everyone else, single mothers' value is \$476. If they are impatient savers at a conventional discount factor, the value depends on how their welfare is counted: \$819 at their own discount factor of 0.70, but \$483 to \$539 at savers' patience, because an impatient saver keeps part of the filing payment and spreads it herself.

The balances favor spenders, by 13 to 29 in log likelihood over an impatient type with a discount factor between 0.50 and 0.90; an impatient type fits nearly as well only at a discount factor far below that range, 0.10, where she barely saves and single mothers' value at savers' patience is \$668 (Appendix Table 26). The balances thus do not separate a mother who keeps nothing from one who keeps almost nothing, and that distinction is worth about a third of the single-mother value (\$987 against \$668). Appendix Figure 8 traces the value against the spender share: from \$476 with no spenders to \$1,824 if every single mother were a spender, and \$844 to \$1,128 over the 95 percent confidence interval of the estimated share. The large single-mother value is therefore conditional on the spender description, which the balances favor. How spenders' consumption is counted matters in one direction only (Table 16): valued with their own impatience, at an annual discount factor of 0.70 or 0.50 or with present bias of 0.70 or 0.50, spenders gain more, and single mothers' average value rises to \$1,027 to \$1,181 (married, \$195 to \$211).

The married average varies less with the description of cash-poor mothers, from \$152 to \$246, than with risk aversion, patience or withholding, which move it from \$112 to \$408; across the rows of Table 15, married cash-poor values vary more, from \$632 to \$1,990. The married spender share is imprecise (2.3 percent, SE 1.5, and 21 percent, SE 14, among married mothers with \$1,000 or less), but the lowest of these values, \$632, comes from the model with savers only: the value of the calendar is concentrated among cash-poor married mothers whether or not any of them are spenders, and the spender share affects only how large it is.

Withholding and permanent payment. Table 16 also relaxes two features of the comparison. The main comparison assumes that the credit reaches a mother only through the IRS. A mother who owes income tax could instead lower her withholding and receive part of the credit in her paychecks. If, under both calendars, mothers lowered their July–December withholding by up to their July–December income tax before credits to receive the part of the credit not yet paid, the values would fall to \$868 for single and \$112 for married mothers. This is an upper bound on what withholding could do: every mother adjusts fully and at no cost, and many families already receive part of the old-law credit through withholding. Three in four married mothers owe enough to receive the remaining half of the credit this way, against one in three single mothers and one in eight cash-poor

single mothers, whose value falls only to $1,652. A permanent policy would also differ from a one-year one, because savers would adjust their buffers over the years. For a mother who starts from the long-run cash distribution under filing-only payment and switches permanently, the 2021 design would be worth $813 a year to the average single mother and $153 to the average married mother. This switch value includes a one-time gain. Under filing-only payment a saver carries the filing payment through the year; under the 2021 design she needs less cash at the start of July, and she consumes the difference once. In the long run that buffer averages $2,474 for married and $2,604 for single mothers valued as savers, 44 and 47 percent of the added credit, and it accounts for almost all of savers' switch value. Comparing the two long-run states instead, each calendar at its own long-run cash distribution, savers gain at most a few dollars a year, and the 2021 design is worth $748 (sampling SE 134) a year to the average single mother and $58 (SE 41) to the average married mother. Spenders, who by construction never build a buffer, value the permanent design at least as much as the one-year one, so in the long run the value of the design comes from the mothers classified as spenders; for married mothers, who are almost all savers, the long-run value is less than a third of the one-year value. Paying the whole credit in twelve monthly payments would be worth $1,433 and $214 a year on switching and $1,347 and $98 in the long-run comparison. These are yearly values of a permanent policy, not one-year spread grants, and they keep each mother's 2021 type probabilities.

Credit. Section 10 shows that access to more card credit lowers the values, mostly for savers.

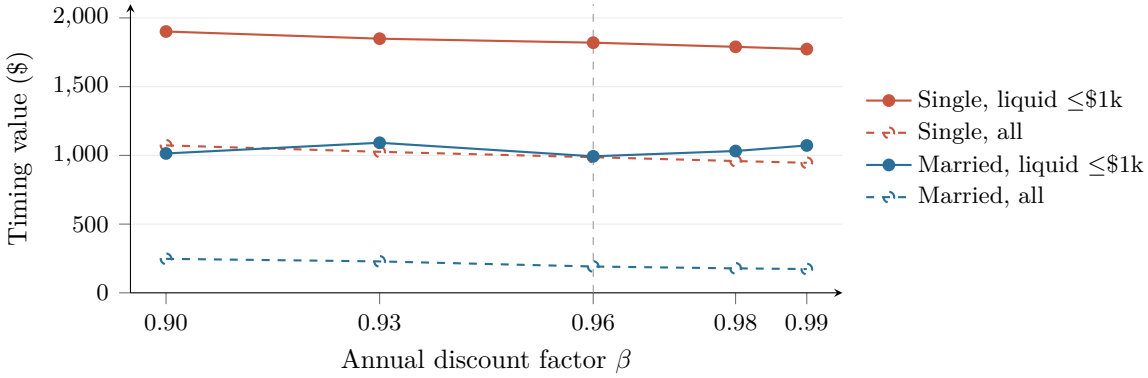


Figure 6: Timing value against the annual discount factor, with the spender model re-estimated at each value. Dashed vertical line: $\beta = 0.96$, the main specification.

9.6 Alternative advance designs

The model can also value calendars that were not used. Table 17 reports the extra value, relative to the actual six advances, of paying the advanced half in one July check, and of advancing the remaining half as well, so that the whole credit arrives by December. A single July check would have been worth *less* than six monthly payments: $22 less for the average married mother and $395 less for the average single mother, because a spender would have consumed it at once (each spender would have lost about $1,350), while savers, who could spread it themselves, would have gained only $7 (married) and $18 (single) from receiving it earlier. Advancing the remaining half would have added value for both, $60 for married and $485 for single mothers, about half of what the actual advances delivered to single mothers. The model values access before filing and the spreading of payments; it is silent on administrative costs and on overpayment risk when income is uncertain, both of which bear on how much to advance.

Table 17: Extra value of alternative advance designs, relative to the 2021 calendar

	All mothers	Liquid ≤$1k	Paid childcare
Married mothers			
Advanced half paid in one July check	−$22 (21)	−$218 (186)	−$10 (24)
Remaining half also advanced	$60 (19)	$372 (157)	$91 (26)
Single mothers			
Advanced half paid in one July check	−$395 (78)	−$822 (135)	−$165 (80)
Remaining half also advanced	$485 (81)	$947 (138)	$432 (126)

Equivalent spread grants under calendars U and M minus calendar S, measured against old-law payment. Standard errors include parameter uncertainty.

10 Credit-card borrowing

The main model rules out borrowing, but more than four in ten married mothers and more than half of single mothers carried card debt at the end of 2020. If mothers could borrow, waiting for the credit would cost them interest rather than consumption, and the value of earlier payment would fall. We extend the model to ask by how much.

10.1 The model with borrowing

Savers can now carry negative cash, which is card debt, at a monthly interest factor $R_b = 1 + 0.1645/12$, where 16.45 percent is the average of the four 2021 quarterly values of the commercial-bank rate on credit-card accounts assessed interest (15.91 to 17.13 percent) (Federal Reserve Board, 2022); positive cash still earns nothing. Like most consumption models with credit cards, the extension tracks a single net position, cash minus card debt, with a higher rate on negative balances (Kaplan and Violante, 2014). It speaks to net liquid holdings and is not designed to explain why many households hold cash and card debt at the same time, the credit card debt puzzle (Gross and Souleles, 2002; Telyukova, 2013; Druedahl and Jørgensen, 2018); Appendix D.7 therefore compares its forecasts with the data on the net position. The budget constraint becomes

$$a_{t+1} = R(\nu_t)\,\nu_t, \qquad \nu_t = a_t + y_t + b_t^X + r_t - f_i d_t - c_t \;\geq\; -L_i/R_b, \qquad R(\nu) = \begin{cases} R_b & \nu < 0 \\ 1 & \nu \geq 0, \end{cases} \tag{9}$$

so the interest rate jumps at zero. Each saver starts from her December 2020 bank balance minus her card debt, and her credit limit is her December 2020 card debt plus k months of resources, $L_i = \text{Debt}_i^{\text{Dec 2020}} + k\,\mu_i$. The SIPP does not record credit limits, so we vary k from zero, where mothers can carry and re-borrow their existing balances but no more, to three months of resources. With $L_i = 0$ the model reduces to the main model, mother by mother. We solve it by the endogenous grid method with the kink at zero, at $\beta = 0.96$.

A spender with the same limit borrows to it in July 2021, consuming the extra credit, and then stays at the limit, paying its interest every month: her consumption falls by $L_i(1 - 1/R_b)$ each month, including in the continuation years. Spenders' values are computed with this debt service, and each mother's values are weighted with her posterior probabilities from Section 7.

Table 18: Timing value with credit-card borrowing

	No	Extra credit beyond December 2020 card debt, k months			
	borrowing	0	0.25	1	3
Married mothers					
All	191 (21)	201 (21)	177 (20)	141 (17)	119 (15)
Liquid assets ≤$1k	993 (141)	980 (141)	892 (135)	750 (125)	636 (115)
Card debt in December 2020	169 (22)	193 (22)	170 (20)	141 (16)	125 (14)
No card debt	207 (32)	207 (32)	182 (30)	142 (27)	114 (24)
Savers, all	152	162	137	100	75
Savers, liquid assets ≤$1k	632	615	519	364	230
Spenders, all	1,132	1,136	1,115	1,080	1,054
Single mothers					
All	987 (118)	1,000 (118)	955 (116)	876 (112)	797 (109)
Liquid assets ≤$1k	1,820 (179)	1,833 (178)	1,771 (177)	1,666 (174)	1,557 (170)
Card debt in December 2020	877 (166)	902 (167)	860 (164)	794 (160)	733 (155)
No card debt	1,125 (163)	1,125 (163)	1,075 (160)	980 (154)	878 (150)
Savers, all	476	490	436	324	184
Savers, liquid assets ≤$1k	761	776	696	522	288
Spenders, all	1,824	1,833	1,799	1,747	1,701

Timing value τ (dollars, equivalent spread grants), posterior-weighted over savers and spenders except in the Savers and Spenders rows, where every mother is valued as that type. Card APR 16.45%, charged monthly at APR/12; cash earns no interest; spenders pay interest on their debt at the limit. Starting position: December 2020 bank balance minus card debt; credit limit: December 2020 card debt plus k months of resources μ_i. Standard errors reflect sampling only, with all parameters, including each mother's posterior spender probability, held at their estimates; they are therefore smaller than the sampling-only standard errors in Appendix Table 22, which re-estimate the spender model in each replicate.

10.2 Results

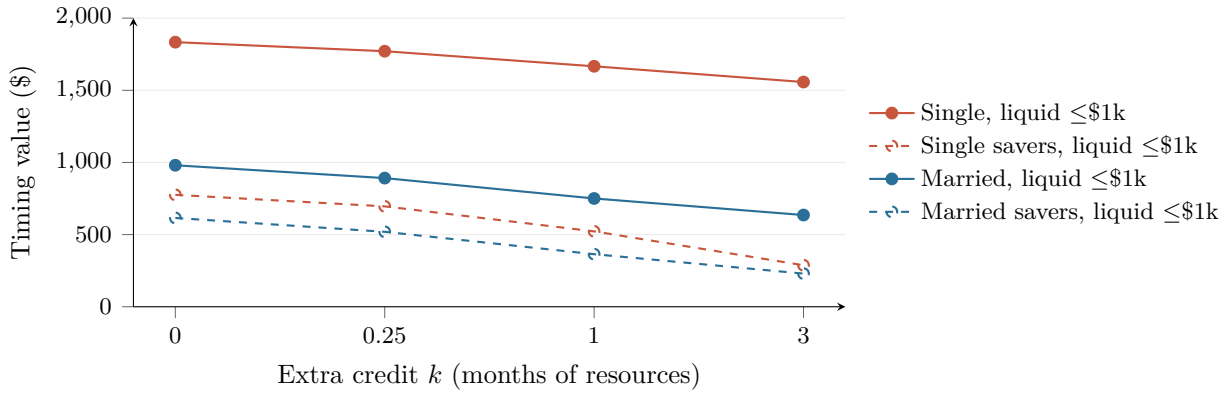


Figure 7: Timing value of mothers with $1,000 or less in liquid assets as access to card credit increases. Card APR 16.45%. Solid: posterior-weighted over savers and spenders; dashed: savers only.

Table 18 and Figure 7 report the results. Three findings stand out.

First, carrying existing card debt can raise the value of earlier payment. With $k = 0$, the timing value of married mothers who carried card debt rises from $169 to $193: an advance pays down card debt, which costs 16 percent a year, months before the filing payment would, so it is worth at least the interest it saves.

Second, new credit substitutes for earlier payment for savers, but at a cost. As the extra credit line grows from zero to three months of resources, the timing value of cash-poor married savers falls from $615 to $230, and that of cash-poor single savers from $776 to $288. The value does not reach zero because borrowing at 16.45 percent to bridge the months until filing is expensive. The decline is largest for mothers without existing card debt, for whom the credit line is new.

Third, credit does much less for spenders. A spender who gains credit consumes it at once and then pays interest on it, so the advances remain her only way to spread the credit over the year; spenders' values fall by about 7 percent as the credit line grows to three months. The values of the mothers with the least cash therefore fall less than savers' values: from $980 to $636 for cash-poor married mothers and from $1,833 to $1,557 for cash-poor single mothers. At every credit limit, cash-poor mothers value earlier payment far more than the average mother, and single mothers more than married mothers.

Because the model nets cash against debt, it understates mothers' use of the advances to repay card debt (Section 8), and mothers who pay card interest while holding cash for other reasons, such as rent that cannot be paid by card, would value earlier payment at least at the interest saved, which this model does not credit them with.

10.3 What the credit did to card debt in the model

The reduced form cannot say whether the credit changed borrowing, because the larger credit, the advances and the stimulus checks arrived together and comparison women differed from mothers in more than the credit (Section 5). The borrowing model can, because it solves each mother's problem under each calendar with everything else held fixed. Table 19 reports savers' expected

Table 19: Model-predicted card debt at the end of December 2021, by calendar

		Savers, $k = 1$ (\$)				$S - O$, by type (\$)			Data (\$)
	N	Debt, O	$F - O$	$S - F$	$S - O$	$k = 0$	$k = 1$	$k = 3$	DiD
Married mothers									
All mothers	593	1,514	103	−517	−414	−233	−392 (33)	−459	−1,909 (681)
>\$1k, card debt	216	2,829	177	−838	−660	−511	−660 (71)	−691	−2,889 (1,222)
>\$1k, no card debt	306	174	28	−129	−101	0	−101 (13)	−149	−1,164 (776)
≤\$1k, card debt	28	7,861	319	−2,143	−1,823	−1,111	−1,596 (237)	−1,803	−2,612 (8,197)
≤\$1k, no card debt	43	1,000	149	−812	−663	0	−471 (43)	−798	805 (909)
Single mothers									
All mothers	147	1,731	145	−789	−644	−202	−388 (61)	−530	−360 (292)
>\$1k, card debt	38	2,086	160	−792	−632	−401	−631 (138)	−744	−1,395 (581)
>\$1k, no card debt	37	231	31	−200	−169	0	−166 (50)	−384	−232 (321)
≤\$1k, card debt	33	3,542	290	−1,513	−1,224	−318	−513 (161)	−615	−401 (832)
≤\$1k, no card debt	39	599	68	−517	−449	0	−134 (36)	−285	991 (438)

Expected card debt (negative cash) at the end of December 2021 from the borrowing model of this section, starting from each mother's December 2020 bank balance minus card debt. Savers: every mother solved as a saver with a credit limit of her December 2020 card debt plus one month of resources ($k = 1$). By type: savers' change weighted by each mother's posterior saver probability; spenders sit at their limit under every calendar and contribute no change. Sampling standard errors in parentheses, with the model held at its estimates. Rows: groups by December 2020 liquid assets (≤ or >\$1,000) and card debt; single mothers: model sample of 147. O: old law; F: expanded credit at filing; S: 2021 calendar. Data (DiD): the difference-in-differences estimate, mothers minus comparison women, coefficients from Table 4 (all mothers) and Table 5, for the change in card debt from December 2020 to December 2021 relative to comparison women, winsorized; single mothers: all 159. No saver has more card debt under S than under O in any group or at any credit limit. The model compares calendars for the same mother; the data compare mothers with comparison women.

card debt at the end of December 2021, after the last advance and before the filing payment, under old-law payment (O), the expanded credit at filing (F) and the 2021 calendar (S). Card debt is not used in estimation, but it comes from the same households and year, so these predictions are a consistency check, not an independent validation.

Three results stand out. First, the larger credit paid only at filing would have *raised* savers' December debt a little, by \$103 for married and \$145 for single mothers valued as savers: a saver who is short of cash and expects a larger payment in April borrows against it. The increase is largest for cash-poor savers who already carried card debt (\$319 and \$290). Second, paying half in advance more than offsets this. Relative to filing-only payment, the advances lower savers' December debt by \$517 and \$789 on average and by about \$1,500 to \$2,100 for cash-poor savers with card debt, and no saver in either sample ends 2021 with more card debt under the 2021 calendar than under old law. Weighted by type, the 2021 calendar lowers December card debt by \$233 to \$459 per married mother and \$202 to \$530 per single mother as the credit line grows from zero to three months. Third, the predictions point the same way as the estimated relative change in card balances for mothers who held cash and carried card debt, but they are smaller: −\$660 and −\$631 at $k = 1$, against −\$2,889 and −\$1,395 relative to comparison women. The gap reflects the netting of cash against debt discussed above, and the two numbers answer different questions. The model does not produce the positive estimates for cash-poor single mothers without card debt; Section 5 shows that these rest on a few large balances. For spenders the model assumes, rather than predicts, the borrowing response, so it says nothing about whether the credit changed their borrowing.

11 Conclusion

The 2021 Child Tax Credit was larger and earlier. The debate about monthly child benefits is about the second feature, so this paper held the amount fixed and asked who was affected by the earlier payments, why, and how much they were worth.

Who. Full refundability paid the most to mothers with low earnings, who also held the least cash; single mothers gained more and held far less. We find no detectable change in their employment rates, paid childcare or bank balances, while their card debt fell relative to comparison women.

Why. The point estimates differed by cash on hand, although few differences between cash groups are statistically distinguishable. The fall in card debt was concentrated among mothers with a buffer and expensive card debt. Among cash-poor single mothers, those who already carried card debt reduced it, and those without it started borrowing at about the same rate as comparison women, 19 percent, and their bank balances showed no detectable relative increase. Low cash was a persistent state for many single mothers. A model with savers and spenders, a parsimonious description that fits the balances better than savers alone, assigns a model-based spender probability averaging 31 percent to single mothers and 2 percent to married mothers. Combined with existing evidence that hardship fell when the payments arrived and rose when they stopped (Parolin et al., 2021; Shafer et al., 2022; Bovell-Ammon et al., 2022), the results are consistent with earlier payment smoothing consumption for mothers who could not do so themselves, although the annual balances cannot show that directly.

How much. A buffer-stock model with income and childcare-cost risk and a group of spenders, with patience set at a standard value (Carroll, 1997) and the other parameters estimated by maximum likelihood on each mother's records, values the 2021 calendar at $191 per married mother and $987 per single mother, relative to paying the same credit at filing, and at $993 and $1,820 for mothers with $1,000 or less in liquid assets. The married average moves with risk aversion and withholding, from $112 to $408, but little with how cash-poor mothers are described. The single-mother values change by up to 12 percent with patience, the starting conditions or work entrants' earnings; they rise with risk aversion, fall by about a third if valued at the advances mothers report receiving, and depend most on how cash-poor mothers are described: if they saved a little, as an impatient saver at a conventional discount factor would, and were valued at savers' patience, the single-mother value would be about half as large. The balances favor spenders, but the large single-mother value should be read as conditional on that description. Extra card credit at market interest rates lowers the values, mostly for savers, since spenders who borrow pay interest on what they consume at once.

Three limitations point to further work. Patience is set rather than estimated, and spenders are valued with savers' discount factor; spenders' values would be lower if part of what they spend on arrival bought durable goods or paid down debt, whose benefits last beyond the month. The single-mother estimates rest on 147 mothers, and the model takes the December 2020 balance as July cash and the 2021 average as the center of monthly income as given. And the model values access to cash and the spreading of payments only: it omits the administrative cost of monthly payments, the risk of overpayment when income is uncertain, and any effect on work, all of which matter for choosing how often to pay.

For policy, the results suggest that paying part of a child benefit before tax time matters most for the families with the least cash, among whom single mothers are overrepresented. Paying the advanced part in monthly pieces matters: a single early check would have been worth less than

the six advances to the mothers who spend what they receive. And single mothers would have gained substantially from advancing more of the credit. Made permanent, the 2021 design would keep most of its value for single mothers in the model, $748 a year when the two long-run states are compared, because spenders by construction never build a buffer; savers, once adjusted, gain almost nothing, so for married mothers only $58 a year remains. The same credit, paid earlier, is worth substantially more to those who cannot wait.

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A Samples, taxes and validation

A.1 One builder for both samples

We build both samples with a single program from the SIPP 2020–2022 releases. To check that the program implements the rules of the original married-mother sample, we first apply it to married mothers. It selects the same 593 mothers, and every model input matches the original inputs exactly: weights and replicate weights, numbers and ages of children, the mother's and spouse's earnings, other income, childcare payments, bank balances and liquid assets in December 2020 and 2021, 2020 family income, and reported advances and stimulus payments. We then apply the same program to single mothers. The rules are:

1. A woman aged 18–54 in December 2021 who is the householder (married mothers: or the householder's spouse), observed in all twelve months of 2021.
2. Marital status: married with spouse present and a single, valid spouse link in every month of 2021, and married in December 2020; or unmarried (widowed, divorced, separated or never married) in every month of 2021 and in December 2020.
3. At least one own child under 18 in the household in every month of 2021 and in December 2020, identified from the children's parent pointers.
4. Valid monthly earnings and family income in all twelve months (missing earnings are set to zero in months of known nonemployment), and, for married mothers, valid spouse earnings in all twelve months.
5. In the universe of the SIPP childcare questions in December 2020 and December 2021, with a known paid-care status.
6. A December 2020 bank record and positive weights in both years.
7. Known advance and stimulus receipt in every month, reported advances no larger than the computed 2021 credit, and positive resources net of childcare under both treatments of other income.

Rules 1–6 give 176 single mothers; rule 7 removes 13 with nonpositive resources after childcare and 4 whose reported advances exceed their computed credit, leaving 159. The model sample of 147 also requires monthly resources μ_i of at least \$500. Comparison women are selected by analogous rules (1, 2, 4 and 6), with no own child under 18 at home in any month of 2021 or in December 2020: 299 married women, as in the original analysis, and 417 single women.

A.2 Taxes and credits

The original married inputs used TAXSIM through the `usincometaxes` R package (Feenberg and Coutts, 1993). We compute taxes for both samples with Tax-Calculator (Policy Simulation Library, 2026) and the same credit formulas: tax before credits and adjusted gross income from Tax-Calculator; the 2021 and old-law CTC from the statutory formulas applied to that income; the earned income tax credit from the statutory schedule, using the smaller of the credits computed on earnings and on adjusted gross income; the child and dependent care credit from Tax-Calculator; and the employee share of payroll taxes, including the additional Medicare tax. Married mothers file jointly and single mothers as heads of household. Table 20 compares the results with the TAXSIM inputs for the 593 married mothers.

Table 20: Tax-Calculator inputs against the original TAXSIM inputs, 593 married mothers

	TAXSIM mean ($)	This paper ($)	Within $5 (%)	Max. gap ($)
Expanded 2021 CTC	5,726	5,726	100.0	0
Prior-law CTC	3,753	3,753	100.0	0
Other refundable credits, 2021	920	917	93.8	127
Resources excluding CTC, 2021	134,141	134,138	93.8	127
EITC, 2021	310	310	100.0	0
Credits under 2022 law	4,259	4,258	98.8	44
Resources under 2022 law	129,702	129,702	100.0	0

A.3 Measurement for single mothers

Three features of single mothers' households matter for interpretation. First, 22 percent live with an unmarried partner. The partner is not part of the SIPP family, so his income is not counted in the mother's resources, although he may share expenses. Second, 23 percent live with other adult relatives, whose income is counted as other family income, as it is for married mothers. Third, single mothers' resources rely more on refundable credits paid at filing: their average 2021 earned income tax credit was $1,505. All refundable credits are paid in April under every calendar, so they raise the value of cash in spring under every calendar alike.

A.4 Mothers and comparison women

Table 21 compares mothers with the comparison women of equation (1) in December 2020. Married mothers and married comparison women have similar incomes and balances; the comparison women are three years older and carried more card debt, which is why the regressions control for starting debt. Single comparison women are better off than single mothers on average, with higher incomes and five times the median liquid assets. Within the group with $1,000 or less, however, single mothers and comparison women have nearly the same median income and bank balance. The remaining differences are age, the larger stimulus payments mothers received for their children, and card debt: cash-poor comparison women carried more, especially those who had any ($7,007 against $4,467 on average). The regressions control for all three, but the larger card balances still shape the lower-income cell of Table 5 (Appendix D).

Table 21: Mothers and comparison women, December 2020

	Married			Single		
	Mothers	Comparison	Difference	Mothers	Comparison	Difference
All women						
Women	593	299		159	417	
Age	38.6	41.5	−2.9 (0.7)	36.6	37.0	−0.4 (0.9)
Income, 2020, median ($)	116,024	118,709	−2,685 (9,140)	28,450	46,200	−17,750 (4,460)
Liquid assets, median ($)	19,000	21,500	−2,500 (4,318)	1,000	5,200	−4,200 (728)
Bank balance, median ($)	15,000	18,600	−3,600 (4,023)	1,000	5,030	−4,030 (763)
Liquid assets ≤$1k (%)	10.8	6.1	4.7 (1.9)	50.7	27.4	23.3 (5.5)
Any card debt (%)	41.0	47.9	−6.9 (4.1)	52.9	45.4	7.5 (5.6)
Card debt, mean ($)	3,732	5,924	−2,192 (834)	2,423	2,970	−547 (622)
Stimulus, 2021, mean ($)	3,722	2,077	1,646 (236)	3,270	1,132	2,137 (304)
Income fell late 2020 (%)	9.9	8.8	1.0 (2.3)	15.9	11.1	4.8 (4.0)
Liquid assets ≤$1k						
Women	71	23		82	113	
Age	36.7	41.6	−4.8 (2.3)	35.6	40.8	−5.2 (1.5)
Income, 2020, median ($)	45,091	48,139	−3,048 (12,606)	24,639	24,624	15 (4,516)
Liquid assets, median ($)	300	255	45 (159)	125	200	−75 (151)
Bank balance, median ($)	230	255	−25 (167)	110	123	−13 (98)
Any card debt (%)	35.6	27.1	8.5 (11.0)	50.6	44.3	6.3 (8.3)
Card debt, mean ($)	3,589	4,707	−1,118 (2,136)	2,262	3,106	−845 (1,131)
Stimulus, 2021, mean ($)	4,070	1,829	2,241 (579)	3,773	1,198	2,575 (490)
Income fell late 2020 (%)	10.5	20.5	−10.1 (9.7)	13.5	14.4	−0.9 (5.9)

Survey-weighted; stimulus: economic impact payments reported for 2021; balances and card debt at the end of December 2020. Comparison women: same marital status, no own children under 18 at home. Difference: mothers minus comparison women, with replicate-weight standard errors. Single mothers: all 159. Income fell late 2020: average monthly family income in September–December 2020 at least 20 percent below January–February 2020.

B Solution, welfare computation and inference

B.1 Solution

For each mother we normalize cash by her monthly resources and solve the saver's problem by the endogenous grid method (Carroll, 2006) on a grid of 600 points for cash. The log-normal income shock is approximated by three-point Gauss–Hermite quadrature (nodes $0, \pm\sqrt{3}\sigma_g$ in logs, weights 2/3 and 1/6), normalized to a mean of one. The continuation year under 2022 law is solved as a stationary twelve-month cycle, iterating to a tolerance of 2×10^{-12} in the consumption policy; its value at July 2022 is the terminal value of the first year, solved backward month by month under each calendar. For a few single mothers with very large fee shares, the stationary iteration does not converge on the default grid and is solved on a grid of 900 points. The spender's value is computed on the same grid and cycle, with consumption equal to cash on hand. Values of the continuation cycle are obtained by solving the linear system for the value of the policy around the cycle.

Each equivalent grant in equation (5) is found by Brent's method, re-solving the old-law problem for each trial grant added to cash in equal parts from July 2021 to June 2022, so that savers re-optimize given the grant. The July-grant measure for savers adds the grant to July 2021 cash. In the model without borrowing a saver's timing value cannot be negative; the most negative computed value is −$1.73, or 0.03 percent of that mother's monthly resources. A spender's timing value can be

negative if the advances arrive in months when her cash is already high; this happens for 3 married mothers.

The borrowing extension adds negative cash to the grid down to the credit limit, with grid points concentrated near the limit and an explicit node at zero, where the interest rate changes. With the credit limit set to zero it reproduces the main model's values mother by mother. The card rate is charged monthly at the APR divided by twelve. If a saver's stationary borrowing problem does not converge within 30 seconds (the consumption policy can cycle at the kink), the code falls back to the next smaller credit line and flags the mother; the aggregation reports the number of such cases. Under the final estimates no mother required the fallback. Spenders' values with debt service use the same continuation cycle as the cash model, with the monthly interest on the limit subtracted from cash in every month after July 2021.

B.2 Likelihood

For each saver, the December 2021 balance contribution is computed from the model's distribution of cash at the end of December 2021 under the 2021 calendar, starting from the mother's own position. The childcare contribution is computed by iterating a four-state system (paying or not, and whether the family has paid yet that year) month by month over 2020 and 2021 from the chain's stationary distribution, which gives the probability of each pair of annual indicators. We store each mother's predictive distribution and evaluate her likelihood on a grid of 161 values of the reporting-error standard deviation between 0.05 and 2.5 (log-spaced), interpolating linearly in its logarithm. Reports of exactly zero, 22 single and 11 married mothers, contribute the probability that the censored report is zero. The mixture parameters are estimated by Nelder–Mead from several starting values.

B.3 Standard errors

The SIPP provides 240 replicate weights for Fay's balanced repeated replication with factor 0.5, so $\widehat{\mathrm{Var}}(\hat{\theta}) = \frac{4}{240}\sum_{r=1}^{240}(\hat{\theta}_r - \hat{\theta})^2$. For each replicate we re-estimate the first-step parameters θ_1 with that replicate's weights and re-fit the savers-and-spenders model. In the re-fit, each mother's saver likelihood and each of her values are linearized in θ_1, $\hat{q}_i(\theta_{1r}) \approx \hat{q}_i(\hat{\theta}_1) + \nabla_{\theta_1}\hat{q}_i\,(\theta_{1r} - \hat{\theta}_1)$, with derivatives from model solutions at $\hat{\theta}_1$ plus and minus a step of 0.02 to 0.04 in each transformed parameter (logs of standard deviations and fee shares, logits of rates). Each replicate's group statistic is computed with that replicate's weights, parameters and posterior probabilities. Sampling-only standard errors hold θ_1 at its estimate. Table 22 shows that first-step uncertainty adds little.

Table 22: Standard errors with and without first-step uncertainty

	Married (593)		Single (147)	
	Full	Sampling only	Full	Sampling only
Spender share (%)	1.5	1.5	5.6	5.4
Reporting-error SD	0.064	0.063	0.192	0.181
Timing value, all ($)	37	36	127	124
Timing value, liquid assets ≤$1k ($)	294	294	203	199

C Employment: CPS details

We use the IPUMS extract of the basic monthly CPS for 2018–2023 (Flood et al., 2024): women aged 18–65, excluding the March ASEC oversample, married with a spouse present or unmarried (separated, divorced, widowed or never married). Mothers are grouped by the number of own children in the household at their first observation, and childless women of the same marital status form the comparison group. Employment is being at work or having a job but not at work. Monthly entry and exit rates come from linking each woman to the previous month. The married sample has 1.26 million person-months from 253,000 women and the single sample 1.13 million person-months from 265,000 women.

For outcome y, women with n own children and calendar month t of 2021–22, the estimand is

$$\Delta_n(t) = \big[y_n(t, 2021/22) - \bar{y}_n(t, 2018\text{–}19)\big] - \big[y_0(t, 2021/22) - \bar{y}_0(t, 2018\text{–}19)\big], \tag{10}$$

where $n = 0$ denotes childless women. Netting each calendar month against its 2018–19 average removes seasonality that differs by number of children; netting against childless women removes shocks common to women of the same marital status. Monthly estimates are averaged into windows: January–June 2021, July–December 2021 (advances), January–February 2022 (no payments), March–April 2022 (filing season), May–December 2022 and 2023. Cell means use the CPS final weight, and standard errors are clustered by person. Table 23 reports the contrasts between windows.

A causal reading requires that, absent the payment changes, the child-group differences would have followed the comparison path. School reopening, the pandemic recovery and other filing-season transfers (notably the earned income tax credit, which is large for single mothers in March and April) need not affect mothers and childless women equally. The filing-season contrasts in particular mix the CTC with these transfers.

Table 23: CPS employment contrasts around the two payment switches

	1 child	2 children	3+ children	All mothers
Married mothers minus married childless women				
Entry into work (pp per month)				
Start: advances minus Jan–Jun 2021	0.20 (0.85)	−1.68 (0.88)	−0.40 (0.93)	−0.64 (0.63)
Stop: Jan–Feb 2022 minus advances	−1.73 (1.23)	0.09 (1.28)	−1.10 (1.34)	−0.92 (0.93)
Refund: Mar–Apr 2022 minus Jan–Feb	4.84 (1.49)	2.48 (1.51)	2.22 (1.57)	3.26 (1.09)
Lump sum minus advances	3.11 (1.25)	2.56 (1.25)	1.12 (1.31)	2.34 (0.90)
Employment (pp)				
Start: advances minus Jan–Jun 2021	0.84 (1.18)	0.12 (1.15)	1.20 (1.43)	0.61 (0.91)
Stop: Jan–Feb 2022 minus advances	−0.68 (1.37)	1.28 (1.32)	−0.86 (1.66)	0.09 (1.06)
Refund: Mar–Apr 2022 minus Jan–Feb	1.79 (1.23)	0.00 (1.20)	1.63 (1.49)	1.01 (0.96)
Lump sum minus advances	1.11 (1.49)	1.28 (1.46)	0.77 (1.83)	1.10 (1.16)
Single mothers minus single childless women				
Entry into work (pp per month)				
Start: advances minus Jan–Jun 2021	0.30 (1.19)	−2.81 (1.64)	2.10 (1.99)	−0.30 (0.96)
Stop: Jan–Feb 2022 minus advances	2.90 (1.83)	6.60 (2.62)	0.91 (3.19)	3.61 (1.48)
Refund: Mar–Apr 2022 minus Jan–Feb	−3.34 (2.07)	−5.04 (3.07)	−1.68 (3.64)	−3.49 (1.70)
Lump sum minus advances	−0.44 (1.64)	1.56 (2.32)	−0.77 (2.84)	0.12 (1.33)
Employment (pp)				
Start: advances minus Jan–Jun 2021	−0.68 (1.25)	−0.10 (1.57)	1.86 (2.05)	−0.14 (0.98)
Stop: Jan–Feb 2022 minus advances	0.65 (1.47)	1.77 (1.84)	0.44 (2.36)	1.00 (1.15)
Refund: Mar–Apr 2022 minus Jan–Feb	1.14 (1.35)	0.74 (1.71)	−1.96 (2.30)	0.47 (1.08)
Lump sum minus advances	1.79 (1.60)	2.51 (2.00)	−1.52 (2.65)	1.47 (1.26)

Percentage points; standard errors clustered by person in parentheses. Each window is netted against the same calendar months in 2018–19 and against childless women of the same marital status. Children counted at first observation.

For married mothers, the start of the advances is not associated with a detectable change in entry or employment in any child group; one- and two-child mothers show higher entry in the filing season than during the advances. For single mothers, the start of the advances is likewise not associated with a change. When the payments stopped, single mothers' entry rate rose by 3.6 points (SE 1.5), and it fell back in the filing season, when single mothers also receive large earned income tax credits; exits rose with entry, and there is no statistically detectable change in employment. The SIPP event studies in Figure 2 agree: for neither married nor single mothers is there a statistically detectable change in employment when the advances began.

D Robustness

D.1 Card debt

Table 24: Change in card debt, mothers minus comparison women: alternative specifications

Specification	Estimate ($)
Married mothers (593 vs 299)	
Main: winsorized 1–99, full controls	−1,909 (681)
No controls	−811 (691)
Starting card debt only	−1,714 (638)
Full controls without the stimulus amount	−1,894 (688)
Not winsorized	−2,041 (762)
Winsorized 5–95	−1,135 (431)
Levels: December 2021 debt, linear control for December 2020 debt	−2,082 (724)
Single mothers (vs 417)	
Main, all 159	−360 (292)
Starting card debt only, all 159	−583 (281)
Main, model sample of 147	−549 (314)
Mothers with card debt in December 2020, all 159	−1,010 (469)
Mothers with card debt in December 2020, model sample of 147	−1,092 (474)

Coefficient on the mother indicator in equation (1). Replicate-weight standard errors. Because December 2020 debt enters linearly as a control, the levels row equals an unwinsorized change regression with that control: it checks the form of the starting-debt control, not a different outcome.

The married estimate depends on controlling for starting debt, because the comparison women carried more card debt at the end of 2020 ($5,924 against $3,732) and high balances tend to fall. With starting debt controlled, the estimates range from $1,135 to $2,082 across specifications.

D.2 Inside the card-debt cells

Table 6 looks inside the cells of Table 5. Panel A reports each group's own balances and changes and, for women who started with card debt, the share who paid it off, or, for those who started without, the share who started borrowing. Panel B reports the mothers-minus-comparison estimate for the main outcome, for the change capped at ±$5,000, for an indicator of any card debt in December 2021 and for the change in the inverse hyperbolic sine of card debt, together with the difference in the paid-off or started share.

Two patterns matter for the text. Among mothers with cash and card debt, the decline survives capping and shows up in the inverse hyperbolic sine in both samples; the share with any debt falls significantly for single mothers, and the paid-off shares point the same way in both samples without being significant at 5 percent. Among cash-poor single mothers, the picture depends on starting debt. Those with card debt cut it at least as much as comparison women: by the measures that do not depend on balance size somewhat more, though not significantly, and with changes capped at $5,000 by $1,001 (SE 499) more. Those without card debt started borrowing at the same rate as comparison women. The positive mean difference comes from the size of new card balances: about ten mothers took on balances with a median of $3,000, against $1,200 among the twelve comparison

women who borrowed, and three of these mothers account for 73 percent of the group's new debt. In the lower-income cell of Table 5, comparison women with card debt started with $8,569 on average, against $2,984 for mothers, and cut it by $4,280, while the mothers' balances rose by $403; women with card debt account for $2,090 of the $2,286 difference in mean changes, almost all of it ($1,962) from the comparison women's paydown. A placebo for the same cells from December 2019 to December 2020 shows no earlier difference (−$586, SE 706, for cash-poor single mothers without card debt), but it rests on 17 mothers. The estimates for cash-poor married mothers rest on 7 and 16 comparison women and are not informative.

D.3 Patience

Table 25: Timing values across discount factors

		Spenders (%)		Married τ ($)		Single τ ($)		Cents lost, all	
β	Log L	Married	Single	All	Liq. ≤$1k	All	Liq. ≤$1k	Married	Single
0.90	−1,243.6	1.3	30.2	247	1,014	1,073	1,901	9.1	41.6
0.93	−1,230.1	2.4	30.1	228	1,092	1,026	1,849	9.0	40.5
0.96 (main)	−1,217.9	2.3	31.1	191	993	987	1,820	7.7	39.9
0.98	−1,211.7	2.9	31.7	178	1,032	959	1,790	7.5	39.4
0.99	−1,209.7	3.4	32.2	173	1,072	946	1,773	7.5	39.2

Timing value τ in dollars (equivalent spread grants). The savers-and-spenders model re-estimated at each β with the first-step parameters fixed; Log L is the December 2021 balance likelihood of the 740 mothers.

Single mothers' values and those of mothers with little cash change little across discount factors; the married average falls from $247 to $173 as savers become more patient. As β rises, savers keep larger buffers and value timing less, but more mothers are classified as spenders. The likelihood of the December balances rises with β over this range (the log likelihood rises by 8.1 from 0.96 to 0.99), so the data do not favor discount factors below the one we use; at $\beta = 0.99$ the average values are 10 percent (married) and 4 percent (single) below those at 0.96.

D.4 An impatient type instead of spenders

Table 26 replaces spenders with an impatient saver whose annual discount factor is set between 0.50 and 0.90, the range used for impatient households in the literature on heterogeneous patience: 0.68 to 0.72 in Aguiar et al. (2025), roughly 0.77 to 0.91 in models matched to liquid wealth (Carroll et al., 2017), and 0.84 to 0.89 in the estimates of Lawrance (1991). We add 0.30 and 0.10, far below that range, to see how the impatient type approaches a spender. Patient savers keep $\beta = 0.96$, and the type probabilities are estimated exactly as for spenders, with intercepts by marital status and a slope on December 2020 cash. Because the impatient type's discount factor differs from the savers', her welfare can be counted in two ways: with her own discount factor, as in that literature, or with the savers' discount factor, as we count spenders.

Table 26: Timing value with an impatient type instead of spenders

		Impatient (%)		Married τ ($)		Single τ ($)		Single, liq. ≤$1k	
β, impatient	Log L	Mar.	Single	Own β	At 0.96	Own β	At 0.96	Own	At 0.96
0.10	−1,219.8	2.3	35.4	246 (64)	164 (18)	1,686 (241)	668 (73)	3,263	1,158
0.30	−1,224.6	2.7	37.9	222 (91)	159 (18)	1,330 (184)	580 (62)	2,518	976
0.50	−1,230.7	3.5	39.6	207 (93)	156 (17)	1,063 (132)	539 (58)	1,966	891
0.60	−1,232.0	6.5	42.1	227 (25)	158 (14)	947 (104)	528 (57)	1,741	870
0.70	−1,235.0	6.5	42.1	205 (22)	157 (14)	819 (88)	514 (56)	1,476	840
0.80	−1,239.5	6.7	42.1	186 (19)	155 (14)	694 (74)	499 (55)	1,215	808
0.86	−1,243.4	6.5	42.1	174 (17)	154 (14)	617 (66)	489 (54)	1,055	788
0.90	−1,246.5	6.6	42.1	166 (16)	153 (14)	564 (61)	483 (53)	945	775
Spenders (main)	−1,217.9	2.3	31.1	–	191	–	987	–	1,820
Savers only	−1,253.5	0	0	–	152	–	476	–	761

Patient savers at $\beta = 0.96$. Each row re-estimates the type probabilities (six parameters, as in the main specification; savers only: three). Impatient: weighted mean prior probability. Timing value τ in dollars (equivalent spread grants), posterior-weighted; "Own β" counts the impatient type's welfare with her own discount factor, "At 0.96" with the savers'. Sampling standard errors in parentheses, re-fitting the type probabilities in each replicate. Rows 0.60 to 0.90: best of 20 starting values (see the text). Spenders and savers only: Tables 12 and 14.

Three results stand out. First, the balances fit worse with any impatient type than with spenders, and worse the more patient the impatient type is: the log likelihood is −1,219.8 at 0.10, −1,230.7 at 0.50 and −1,246.5 at 0.90, against −1,217.9 with spenders, with the same number of parameters. The fit approaches that of spenders only as the impatient type nearly stops saving: at 0.10 she discounts the next month by 17 percent. This is the pattern that led us to spenders: the balances keep pointing to mothers who hold nothing rather than a little. Between 35 and 42 percent of single mothers and 2 to 7 percent of married mothers are classified as impatient. For impatient discount factors of 0.60 to 0.90 the likelihood is flat in the type probabilities; the rows report the best fit from 20 starting values, which improves on the starting values of the original estimation by up to 1.8 and makes the type probability a step function of December 2020 cash. At these fits the married impatient share is 6.5 to 6.7 percent at every discount factor from 0.60 to 0.90. Re-fitting the main estimate from 16 random starting values raises its log likelihood by at most 0.03.

Second, counted with their own discount factors, impatient mothers value the 2021 calendar highly, because they value money sooner for its own sake. Single mothers' average value ranges from $564 at 0.90 to $1,063 at 0.50, and is $819 at 0.70, close to the impatient discount factors in Aguiar et al. (2025); the spender model's $987 lies inside this range; at 0.30 and 0.10 the single-mother value is $1,330 and $1,686. Married mothers' values range from $166 to $227 between 0.50 and 0.90, around the main value of $191, and are $222 and $246 at 0.30 and 0.10.

Third, counted with the savers' discount factor, impatient mothers gain much less, because they save part of the filing payment and spread it themselves: single mothers' average value is $483 to $539, only slightly above the value for savers alone ($476), and married mothers' $153 to $158. As the impatient type comes closer to spending everything, her value at the savers' discount factor rises toward the spender value: $580 at 0.30 and $668 at 0.10 for single mothers, against $987 with spenders. The size of the single-mother value in the main specification therefore rests on two things: what the balances show, that about a third of single mothers keep nothing rather than a little, and counting the smoothing of their consumption at patient rates. With patient savers at $\beta = 0.98$ the pattern is the same: the log likelihood is −1,224.7 to −1,240.4 with an impatient type

between 0.50 and 0.90 (−1,218.4 and −1,213.4 at 0.30 and 0.10) against −1,211.7 with spenders, and single mothers' average value is $541 to $1,055 with own discount factors and $460 to $516 at the savers' discount factor ($558 and $647 at 0.30 and 0.10).

D.5 The spender share and the value

Figure 8 shows how the average timing value depends on the share of spenders. We shift both marital-status intercepts of the spender prior by a common amount, holding the slope on starting cash and the reporting error at their estimates, recompute each mother's posterior spender probability given her December 2021 balance, and average her saver and spender values with it. The left end of each curve is the value with savers only, the right end the value if every mother were a spender. For all single mothers the value rises from $476 to $1,824; over the 95 percent confidence interval of the estimated share, 20 to 42 percent, it ranges from $844 to $1,128. For married mothers the estimate lies near the savers-only end, and over its interval, 0 to 5 percent, the value ranges from $152 to $235. The curves are concave, because the mothers classified as spenders first, those with December 2021 balances near zero, are those whose spender and saver values differ most: among mothers with $100 or less in December 2021, the spender value exceeds the saver value by $1,623 for single and $1,769 for married mothers, against $1,266 and $948 for the others.

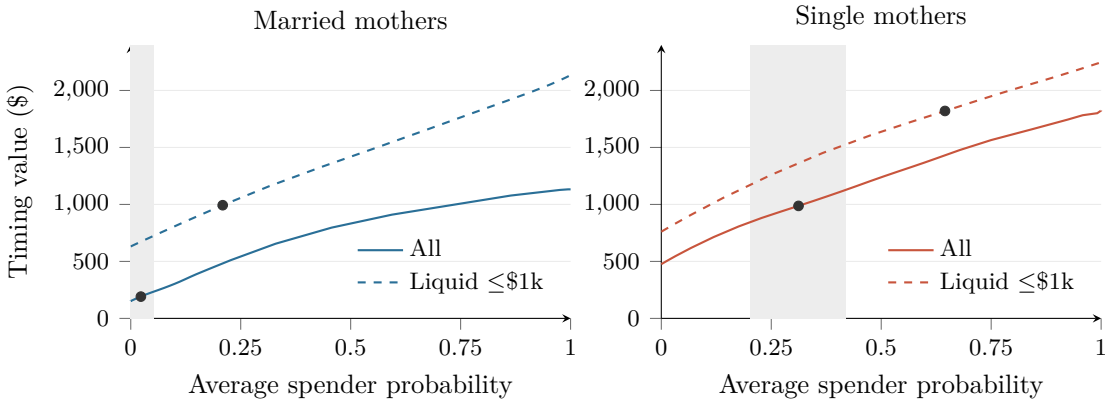


Figure 8: Timing value against the average spender probability. Both intercepts of the spender prior are shifted by a common amount, with the other estimates fixed; each mother's saver and spender values are weighted by her posterior spender probability. Left end of each curve: savers only; right end: every mother a spender. Dots: the estimates. Shaded: 95% confidence interval of the estimated spender share of all mothers in the sample.

D.6 Liquid assets of married mothers

If the advances were saved, liquid assets should have grown relative to comparison women. For married mothers we find no detectable difference in the change in log liquid assets from December 2020 to December 2021 (0.09, SE 0.11), or in the share with $1,000 or less. The mean dollar difference is driven by mothers with $25,000 or more and by stock holdings (married mothers became 9.7 points, SE 4.1, more likely than comparison women to hold stocks or mutual funds), while the median change differs by only $1,300 (SE 2,300); the 2019–2020 change shows no difference.

D.7 Out of sample: December 2022 balances

The 2023 release of the SIPP reports bank balances at the end of December 2022 for mothers in the 2020 and 2021 panels. We find 424 of the 593 married and 96 of the 147 single mothers in the model samples. The public-use release gives the 2022 balances by person, and we sum them within the household; in the 2020 release, which has both, this sum reproduces the household total for 98 percent of households. December 2022 comes after the April 2022 filing payment, which paid the remaining half of the expanded credit, and after six months under 2022 law. We forecast each mother's December 2022 balance from the model as estimated, without re-estimating anything: a saver follows the policies of the 2021 calendar through June 2022 and then the 2022-law cycle, and a spender holds nothing. Each mother's type weight is her posterior spender probability given her December 2021 balance, so the forecast uses information through December 2021 only. The mothers found in 2023 have average posterior spender probabilities close to those of the full samples (2.6 and 30.1 percent, against 2.3 and 31.2).

Table 27: December 2022 bank balances: data and out-of-sample forecast

		Observed	Model	Model, reported	Savers only	December 2021
Married mothers						
All (424)	Mean ($)	64,088	36,724	125,829	36,846	59,574
	Median ($)	16,200	9,923	10,234	10,194	17,300
	Share ≤$1,000 (%)	10.6	6.4	12.4	4.0	9.0
Liquid ≤$1k (49)	Mean ($)	6,627	3,529	12,616	4,727	5,463
	Median ($)	1,250	2,516	2,130	3,767	1,000
	Share ≤$1,000 (%)	47.8	32.6	36.3	9.3	52.2
Single mothers						
All (96)	Mean ($)	10,003	7,212	22,489	8,401	11,188
	Median ($)	1,512	3,037	2,683	4,427	2,000
	Share ≤$1,000 (%)	42.7	34.1	35.2	5.7	41.7
Liquid ≤$1k (47)	Mean ($)	3,740	1,746	6,152	4,151	3,026
	Median ($)	800	0	533	3,667	500
	Share ≤$1,000 (%)	55.5	64.1	59.8	6.9	62.7

Mothers in the model samples found in the 2023 SIPP; December 2020 liquid assets define the groups; 2022 survey weights. Model: each mother's forecast distribution of her December 2022 bank balance, a mixture of her saver forecast and a spender's zero balance with her posterior spender probability given the December 2021 balance, all parameters at the main estimates. Model, reported: the same with the estimated reporting error. Savers only: the saver forecasts alone. December 2021: each mother's own balance a year earlier. Same format as Table 11.

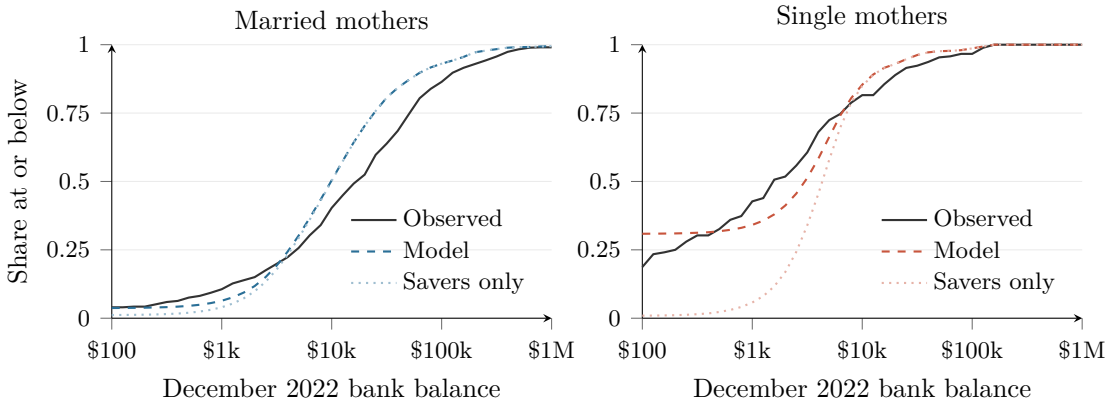


Figure 9: Cumulative distribution of December 2022 bank balances, mothers found in the 2023 SIPP: data and the model's out-of-sample forecast (log scale), survey-weighted. Model: each mother's saver forecast mixed with a spender's zero balance using her posterior spender probability given the December 2021 balance, without reporting error. Savers only: every mother forecast as a saver.

Three results stand out. First, spenders are what make the forecast work. With the estimated reporting error, the model forecasts that 60 percent of cash-poor single mothers report $1,000 or less in December 2022, against 55 percent observed; if every mother were a saver, it would forecast 16 percent (7 percent without reporting error, as in the figure), because a saver keeps part of the April filing payment. The naive forecast that each mother's balance stays at its December 2021 level is about as close (63 percent), as it should be if spenders hold nothing through the filing payment; the check therefore speaks to levels of balances and to the spender description, not to the value of earlier payment. Second, the forecast for cash-poor married mothers is too low, 36 against 48 percent (22 percent with savers only, 9 percent without reporting error; the naive forecast is 52 percent), as the fit for December 2021 was in Table 11: more of them keep little cash than the estimated spender share implies. Third, as in December 2021, the model understates the balances of mothers with more cash, and it does not anticipate that single mothers with more than $1,000 in December 2020 drew down their balances during 2022: 30 percent of them ended 2022 with $1,000 or less, against a forecast of 11 percent. The model has no common shock in 2022, a year of high inflation.

The borrowing extension of Section 10 tracks a single state, the net position: cash minus card debt. It cannot split that position into cash and debt, so we compare it with the data on the net position itself. Table 28 reports, for the same mothers, the share whose household card debt exceeded their bank balance at the end of December 2022, and the model's forecast with spenders' credit limits set at their December 2020 card debt ($k = 0$) or one month of resources more ($k = 1$). The forecast is close for all mothers: 16 percent of married mothers ended 2022 with more card debt than cash, against forecasts of 8 and 19 percent, and 32 percent of single mothers, against 29 and 47 percent. Two groups stand out. Cash-poor mothers without card debt rarely went into net debt (10 percent of married and 9 percent of single mothers), as the model predicts if spenders take on no new credit ($k = 0$), but not if they can borrow another month of resources. Cash-poor single mothers with card debt were in net debt less often than forecast (45 against 68 and 73 percent): many of them paid debt down, which a spender at her limit does not do. In levels, the model understates card debt, because it nets cash against debt while many mothers hold both: with $k = 1$ it forecasts that 22 percent of married mothers who had more than $1,000 and card debt in December 2020 still carry card debt in December 2022, against 75 percent in the data. This is the limitation noted in

Section 8 for the December 2021 decline in card debt.

Table 28: Share of mothers with more card debt than cash, December 2022: data and forecast

			Model		Savers only
	N	Observed (%)	$k = 0$	$k = 1$	$k = 1$
Married mothers					
All	424	16	8	19	17
>\$1k, card debt	155	27	18	22	22
>\$1k, no card debt	220	7	0	9	9
≤\$1k, card debt	18	48	52	59	55
≤\$1k, no card debt	31	10	0	49	24
Single mothers					
All	96	32	29	47	25
>\$1k, card debt	23	41	18	25	25
>\$1k, no card debt	26	18	0	12	12
≤\$1k, card debt	27	45	68	73	39
≤\$1k, no card debt	20	9	0	80	13

Mothers in the model samples found in the 2023 SIPP; 2022 survey weights. Share with a net position (household bank balance minus household card debt) below zero at the end of December 2022. Model: forecast from the borrowing extension of Section 10, without reporting error: savers start from their December 2020 bank balance minus card debt, with a credit limit of their December 2020 card debt plus k months of resources, and follow the 2021 calendar through June 2022 and the 2022-law cycle after; spenders sit at their limit; type weights are posterior spender probabilities given the December 2021 bank balance, all parameters at the main estimates. Rows: groups by December 2020 liquid assets (≤ or >\$1,000) and card debt.